
Estimating Active Dimension of Training Dynamics from One Scalar Log, with an Application to Grokking

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Abstract

1 Over a stretch of training in which the update rule is fixed and the parameters keep
2 returning near states already visited, the states they revisit form a set of dimension
3 far below the number of parameters: the active dimension of that regime. Measuring
4 it needs the stored trajectory, whereas a run already writes scalar logs: how much of
5 that dimension survives in one such scalar? We reconstruct a state space by delay
6 coordinates and estimate its intrinsic dimension by maximum likelihood on nearest-
7 neighbour distances. The procedure returns a plausible number on any record,
8 including one in which nothing recurs, so it estimates the active dimension only
9 under conditions we establish separately. On the one of six constructed systems that
10 meets every requirement we fixed in advance it is accurate to about one component,
11 up to a ceiling near eight; The remaining constructions expose failure modes under
12 transience, stochastic forcing and poor delay choices. All validation systems are
13 externally forced; the training runs are not. The diagnostics therefore reject the
14 scalar estimate as a dimension in both grokking settings. We instead analyse
15 saved trajectory sketches. The detrended covariance participation ratio falls near
16 generalisation in four regularised runs and later recovers. A memorising run can
17 fall as far. The timing of the fall and its reversal separates the generalising runs
18 in this sample. A matched short-window analysis of the parameter norm shows
19 the same event, but its level is not dimensional and the effect is weaker than the
20 corresponding change in roughness.

21 1 Introduction

22 A network is trained in a parameter space of dimension P , but over a stretch of training the parameters
23 usually move in far fewer dimensions. Is an apparent simplification of training a real reduction in
24 the number of degrees of freedom in use, or an artefact of a shrinking scale? Can a phase transition
25 such as delayed generalisation be detected without instrumenting the run? Both are argued from a
26 count of degrees of freedom. Those in use disagree: the *available* dimension of the subspace the
27 parameters are confined to, which Li et al. [2018] take as the smallest that still solves the task, the
28 *functional* dimension of the map from that subspace to the outputs, and the *effective rank* of the
29 trajectory covariance.

30 Our two questions concern a fourth quantity: the number of independent components of the set the
31 parameters occupy while the update rule is fixed. We call it the *active dimension* of that regime
32 (equation (1)). If the motion is forced by r independent quasiperiodic phases, the path lies on an
33 r -dimensional torus and the active dimension is r . Section 5.1 moves it while the available and
34 functional dimensions stay fixed.

35 Measuring the active dimension means storing the parameter trajectory, which an ordinary run does
36 not do. A run already writes scalar logs: a loss, a gradient norm, a parameter norm. *How much of the*
37 *active dimension can be recovered from one such scalar?* Nonlinear time series analysis supplies a

standard answer: reconstruct a state space by delay coordinates [Takens, 1981, Sauer et al., 1991], then estimate its intrinsic dimension by maximum likelihood on nearest-neighbour distances [Levina and Bickel, 2004, MacKay and Ghahramani, 2005].

The procedure returns a number on any record. Power-law noise yields a finite and reproducible value fixed by its spectral exponent [Osborne and Provenzale, 1989, Theiler, 1991]; a decaying transient yields a large value. Establishing which case a training log falls in is the substance of the problem.

Contributions. This paper makes three contributions. First, it defines active dimension as a local dimension of the occupation measure of a recurrent training regime and separates it from available dimension, functional dimension and trajectory effective rank. Second, it tests scalar-log recovery on six constructed systems under a frozen, held-out protocol. One learned-system benchmark passes every prespecified check; the other systems expose failure modes caused by transient motion, stochastic forcing, unequal excitation and poor delay choices. Third, it applies the same diagnostics to two grokking settings. The scalar estimate is not dimensional there. Saved trajectory sketches instead reveal a reversible, regularisation-associated drop in detrended covariance participation ratio near generalisation in four runs, while matched short-window logs provide a transition- correlated signal with no dimensional interpretation.

2 Related work

Reconstruction and intrinsic dimension. Delay reconstruction follows from Takens’ theorem and its extensions (Appendix A), is used on empirical series [Sugihara and May, 1990, Sugihara et al., 2012], and has standard rules for choosing its lag and dimension [Fraser and Swinney, 1986, Kennel et al., 1992, Cao, 1997, Broomhead and King, 1986]. We use pooled maximum likelihood [Levina and Bickel, 2004, MacKay and Ghahramani, 2005]; the alternatives [Grassberger and Procaccia, 1983, Facco et al., 2017, Camastra and Staiano, 2016] are bounded alike by a finite record [Eckmann and Ruelle, 1992]. Oversampling [Theiler, 1986] and power-law noise [Osborne and Provenzale, 1989] produce spurious dimensions; surrogate tests [Theiler et al., 1992, Schreiber and Schmitz, 1996] detect nonlinear structure but not how much is present and miss a decaying transient entirely.

Dimension in optimisation. Random-subspace training [Li et al., 2018, Aghajanyan et al., 2021] reports the smallest k that still solves a task, a property of the task and not of a window; we adopt the same parameterisation but measure a different quantity within it. Closest to the active dimension is the finding that descent occurs largely in a small subspace spanned by the top Hessian eigenvectors [Gur-Ari et al., 2018, Sagun et al., 2017]. Ansuini et al. [2019] and Pope et al. [2021] estimate the intrinsic dimension of a data set or of a layer’s representations, properties of a point cloud rather than of a path. The participation ratio we adopt [Gao et al., 2017] is not the spectral-entropy quantity of the same name [Roy and Vetterli, 2007]; it has been measured on a grokking trajectory once, in passing [Notsawo Jr. et al., 2023]. All of these quantities need weights, gradients or activations, whereas the nearest precedent for a scalar computed during training and acted on is the gradient noise scale [McCandlish et al., 2018].

Grokking. Delayed generalisation [Power et al., 2022] is studied in two settings and we use both. The transformer setting is regularised and, in our version, stochastic [Nanda et al., 2023, Liu et al., 2023]; the perceptron setting is the unregularised full-batch gradient descent of Gromov [2023], on the modular polynomials and representability criterion of Doshi et al. [2024], which supply label-matched pairs (Appendix J). Explanations are framed in weight or activation space [Nanda et al., 2023, Varma et al., 2023, Liu et al., 2022, Barak et al., 2022] or in a lazy-to-rich transition [Kumar et al., 2024]. Closest to us, Notsawo Jr. et al. [2023] predict grokking from early loss oscillations [Thilak et al., 2022] and Merrill et al. [2023] track competing subnetworks. We differ from all of these in estimating a dynamical quantity whose ground truth is measured independently.

84 3 Problem statement

85 3.1 The active dimension

86 Let $\mathbf{s}_t = (\boldsymbol{\theta}_t, \mathbf{u}_t)$ be the optimiser state at step t : the parameter vector $\boldsymbol{\theta}_t \in \mathbb{R}^P$ and whatever the
 87 optimiser carries beside it. Where the parameters are confined by construction or by the dynamics to
 88 an affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{V}^\top \mathbf{c}$ with $\mathbf{V} \in \mathbb{R}^{k \times P}$ orthonormal, we work in the free coordinates
 89 $\mathbf{c}_t \in \mathbb{R}^k$; that change of coordinates is an isometry and preserves intrinsic dimension.

90 We call k the *available* dimension. The *functional* dimension is the rank of $\partial \mathbf{f}(\mathbf{c}) / \partial \mathbf{c}$, with $\mathbf{f}$ the
 91 outputs on a fixed probe set; it bounds how many of the k directions can change the computed
 92 function. The *active* dimension, the subject of this paper, is a property of a *regime* R : a stretch of
 93 training over which the update rule is fixed and the state carries an ergodic invariant measure. Carried
 94 to the free coordinates, that measure gives the *occupation measure* μ_R , the long-run fraction of steps
 95 at which $\mathbf{c}_t$ falls in a given set. The regime occupies the support $\mathcal{A}_R$ of μ_R , the points whose every
 96 neighbourhood is visited with positive frequency. The active dimension is the exponent with which
 97 μ_R fills a small ball about such a point,

$$d_{\text{act}}(R) = \lim_{\varepsilon \downarrow 0} \frac{\log \mu_R(B(\mathbf{c}, \varepsilon))}{\log \varepsilon}, \quad (1)$$

98 with $B(\mathbf{c}, \varepsilon)$ the ball of radius ε about $\mathbf{c}$. A measure whose exponent is the same at μ_R -almost
 99 every $\mathbf{c}$ is *exactly dimensional*; that common value is $d_{\text{act}}(R)$. Where $\mathcal{A}_R$ is a smooth d -dimensional
 100 manifold carrying a positive density the exponent is d ; a point of it is then fixed by d local coordinates,
 101 the *active components*.

102 No record reaches the limit. Between two radii the same ratio gives the finite-scale exponent
 103 $d_{\text{act}}(R; \varepsilon_1, \varepsilon_2)$ of equation (6), which agrees with $d_{\text{act}}(R)$ where the same power continues below ε_1
 104 and otherwise counts only the phases forced above it (section 6.3). A window $\mathcal{W}$ of L steps samples
 105 μ_R rather than being the set: its own exponent is zero, and it returns that finite-scale exponent at the
 106 radii its neighbour distances span.

107 The systems of section 5 set $\mathbf{c}_t = F(\varphi_1(t), \dots, \varphi_r(t))$ with r rationally independent quasiperiodic
 108 phases and F a smooth embedding of the r -torus, giving $\mathcal{A}_R = F(\mathbb{T}^r)$ and $d_{\text{act}}(R) = r$. Rational
 109 independence makes the orbit dense on the torus; the embedding keeps F from merging or discarding
 110 a phase.

111 We also measure the participation ratio [Gao et al., 2017], which for a non-negative spectrum
 112 $\lambda_1, \dots, \lambda_n$ is

$$\text{PR}(\lambda) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (2)$$

113 It equals m exactly when m of the λ_i are equal and the rest vanish. Applied over a window to the
 114 covariance of $\mathbf{c}_t$ it gives the *effective rank* PR^{pos} and to the covariance of the residual $\tilde{\mathbf{c}}_t$ about the
 115 window's least-squares line the detrended form PR^{det} . A drifting trajectory needs the detrended
 116 form, since a straight drift dominates the spectrum and holds PR^{pos} near unity however the trajectory
 117 moves about that line. Written without a superscript, the *effective rank* means PR^{pos} .

118 The effective rank is not the active dimension: it falls below r wherever the r directions do not carry
 119 comparable variance, while $d_{\text{act}}(R)$ stays at r (section 6.3).

120 3.2 Observation model and estimator

121 We observe a scalar series $x_t = \phi(\mathbf{s}_t)$, in which $\mathbf{s}_t$ is the full optimiser state and ϕ is fixed before the
 122 run. From it we reconstruct $\mathbf{y}_t = (x_t, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and
 123 observation on a compact invariant set of box-counting dimension d , this map is an embedding once
 124 $E > 2d$ (theorems 1 and 2). Genericity is not verifiable for a ϕ fixed in advance, so every result below
 125 is reported over a family of observers. We estimate the intrinsic dimension of $\{\mathbf{y}_t\}$ by maximum
 126 likelihood on nearest-neighbour distances (theorem 4). Let $r_1^{(i)} \leq \dots \leq r_m^{(i)}$ be the distances from
 127 point i to its m nearest neighbours, counting a neighbour only if it is more than W_T steps away from i
 128 in time (figure 1). This is the *Theiler exclusion* [Theiler, 1986, Kantz and Schreiber, 2004]: without it
 129 the nearest points in $\mathbb{R}^E$ are the samples just before and after i , so the statistic measures the sampling

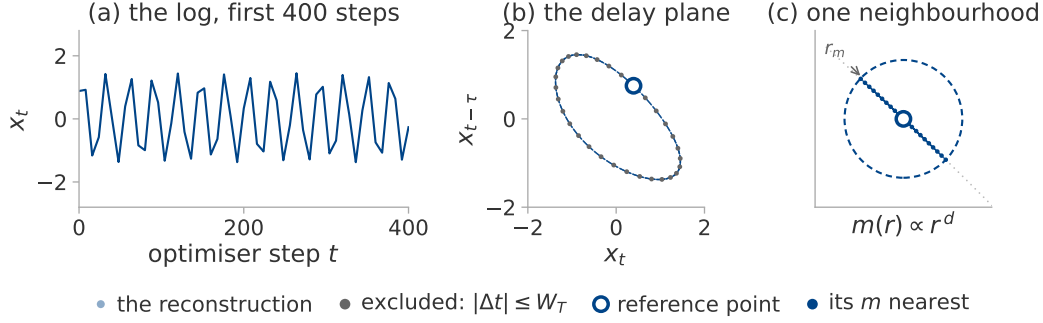


Figure 1: One scalar log carries the geometry the estimator counts. The one-phase case of section 5.1: (a) 400 of its 10 000 logged steps; (b) the reconstruction of the whole record in its first two delay coordinates, a closed loop; (c) the neighbours of one point and the radius r_m of equation (3). The record passes round the loop many times over, so the excluded samples lie all round it and every neighbour in (c) is a return from another pass.

rate instead of the geometry. Pooling the likelihood over the N points before inverting, as MacKay and Ghahramani [2005] recommend, gives

$$\hat{d}_{\text{MG}}^{-1} = \frac{1}{N(m-1)} \sum_{i=1}^N \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}, \quad (3)$$

Appendix B.1 gives the algorithm as implemented; table 8 scores the per-point average of Levina and Bickel [2004] beside it. The *roughness* $\text{std}(\Delta x)/\text{std}(x)$, the *linear participation ratio* PR_{delay} of equation (2) on $\text{Cov}\{\mathbf{y}_t\}$ and the *spectral participation ratio* on the power spectrum of x accompany every estimate as nulls: each can grow with r for reasons that involve no geometry, so an estimator no more accurate than the best of them has measured nothing.

3.3 Dynamical regimes

Before an estimate computed from a log gives an active dimension, the window must lie inside a single regime. The embedding theorems (theorems 1 and 2) require a deterministic map on a compact invariant set, which licenses the delay reconstruction as a change of coordinates. The neighbour statistic of equation (3) requires more: the record must cover that set densely enough that the Theiler exclusion still leaves each point a genuine return. Recurrence is needed because our entire sample is one trajectory. Last, the neighbour radii must span a range over which the exponent is constant. Where a condition fails, $\hat{d}_{\text{MG}}$ is a statistic of the window and not a dimension.

- **Deterministic and recurrent.** The path returns close to states already passed through and lies on an r -dimensional torus, so $\mathcal{A}_R$ has r independent components and $d_{\text{act}}(R) = r$; the estimator recovers it.
- **Deterministic and transient.** A converging run decays along a curve it never revisits. The curve is one-dimensional, but a run that does not return samples no invariant measure, so it has no $\mathcal{A}_R$ and equation (1) has nothing to measure: the dimension of a path and the dimension of a regime are different quantities, and here the estimator returns a statistic of the window (section 6.1). Its effective rank does stay near unity (table 11), because the curve is nearly straight.
- **Stochastically forced.** A stationary mini-batch optimiser may have an occupation measure of its own, whose dimension counts the directions the gradient noise excites. That is not the quantity we are after. The embedding theorems for forced systems (theorem 3) do cover forcing of this kind, but one realisation of it at a time, whereas a single run crosses realisations.

3.4 Diagnostics

The series alone is enough to tell a transient from stochastic forcing. The **identifiability ratio** $\rho_{\text{ident}} = \hat{d}_{\text{MG}}(2E_{\text{max}})/\hat{d}_{\text{MG}}(E_{\text{max}})$ compares the estimate at a doubled embedding dimension with the estimate at E_{max} , the one a configuration fixes. The ratio stays near unity where a dimension can be measured and grows wherever the estimate follows the embedding dimension rather than the data:

under mini-batch noise, and furthest of all on a transient (figure 2b). It does not say which of the two has happened. The **trend-crossing count** does, being the number of sign changes of the residual of the window about its own least-squares line, near zero on a monotone decay. Neither tests recurrence itself (Appendix I); nothing we compute tests the scaling range.

4 Validation protocol

The requirements below were fixed before any result was computed; table 4 records which of them each system fails.

Constructed truth: the ground truth follows from the construction rather than from a fit. Every system is built with r independent quasiperiodic phases, so that $d_{\text{act}}(R) = r$; the effective rank of the recorded trajectory must then confirm that all r are comparably excited.

Frozen configuration: the estimator is chosen once, by a search over its own settings alone. Searching over the system at the same time would select the data together with the method.

Withheld ranks: ranks are held out as well as seeds, because the geometry of the excitation depends on r and every seed at a given rank shares that geometry.

Zero learning rate: every system is re-run with the learning rate set to zero, so any observer of the optimiser must be constant. A log that still varies and still returns its earlier estimate was never a function of the optimiser.

5 Evaluation on systems with a known active dimension

We evaluate the estimator on six systems of increasing realism, from an oscillating diagonal matrix to a linear head on a network trained on image data (table 4). The first has no optimiser and no feedback: in $\mathbf{W}(t) = \text{diag}(b_i + \delta_i(t))$ the offsets b_i are fixed and r of the δ_i oscillate at rationally independent frequencies; the estimate tracks r across the whole withheld range to within a third of a component.

5.1 A constrained head on a network trained on image data

A tanh network is trained on the scikit-learn handwritten digits and frozen; a linear head confined to a k -dimensional affine subspace is then trained on top of it under softmax cross-entropy, so the curvature is data-dependent and all the dynamics occur in c . The excitation is made r -dimensional by modulating the loss weights of r of twelve fixed data groups (Appendix E). We force the head quasiperiodically, with mini-batch noise, and by plain gradient descent, covering the recurrent, stochastic and transient regimes of section 3.3.

Twelve scalar observers are recorded (Appendix B.2). The estimator was frozen on four of them, on seeds and ranks reserved for that purpose, so every result below is out of sample in rank, in seed, and for most observers in the observer too. In the deterministic recurrent regime the estimate recovers the active dimension to within about one component. Every observer orders the withheld ranks correctly, and none does so through the roughness of its log (table 6).

The ceiling. At the frozen eight-component configuration the estimator saturates above about eight components; at $k = 20$, under the twenty-component one, it tracks the rank to about ten and then falls behind (table 7). Over every configuration we swept the largest r it tracks follows neither the embedding dimension nor the record length and never exceeds about eleven (Appendix L). Most of it is the number of components the delay window resolves linearly: a linear participation ratio on the same covariance saturates along the same curve.

The zero-learning-rate test. The check of section 4 invalidated four of our six systems and the function-subspace variant besides: each returns its reported estimate with the parameters frozen, because the forcing keeps the residual moving while the parameters do not. It also rejects the instantaneous mini-batch loss of this system, otherwise among the more accurate observers.

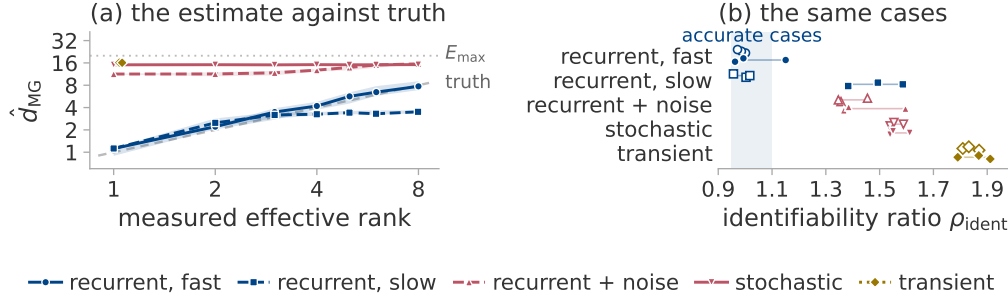


Figure 2: Recovery of the active dimension in each regime. The linear head of section 5.1 under five excitations, coloured by the regime of section 3.3: two quasiperiodic forcings at different rates, the fast one with gradient noise added, mini-batch noise alone, and plain gradient descent. (a) Estimate against truth: marker, median over seeds and observers; band, their interquartile range; dashed, exact recovery; dotted, $E_{\max}$. (b) The identifiability ratio, one point per observer, at $r = 2$ (open) and $r = 6$ (solid).

6 Conditions of validity

6.1 Recovery in each regime

In figure 2(a) the fast quasiperiodic forcing follows the diagonal over the whole withheld range. Under mini-batch noise the estimate is flat in r and well above the truth, responding to $E_{\max}$ and to the length of the record instead of to the data. Noise at a fortieth of the forcing amplitude leaves the ordering nearly intact; it still moves the estimate at $r = 1$ from 1.1 to 11.3. The slow forcing changes behaviour partway along: it tracks the truth to about $r = 4$, then saturates.

Panel (b) plots the identifiability ratio on either side of that change. The regime labels alone would have predicted the extreme cases, not the slow forcing, which passes at $r = 2$ and fails at $r = 6$. The ratio therefore locates the failure inside a single system, using nothing the ground truth supplied.

The Theiler exclusion is why the estimator errs most on a transient: the 16 it returns there is a divergent quantity evaluated at the exclusion the configuration asks for. Removing it returns the value maximum likelihood gives on a uniformly sampled curve, at every rank, seed and observer, and leaves the fast recurrent case unchanged (Appendix I.2).

Every validation system above is externally forced. Training is not. The evaluation therefore establishes a conditional result. Full-batch descent does reach the edge of stability [Cohen et al., 2021], where an unforced run might still recur, but oscillation is not recurrence and the diagnostics reject every run at the edge (Appendix K).

6.2 The delay lag

The delay window spans $(E_{\max} - 1)\tau$ samples. Below an appreciable fraction of the oscillation period the delay coordinates are nearly collinear and the torus never unfolds [Caspdagli et al., 1991]; varying τ alone on a fixed torus moves the estimate over more than an order of magnitude, through the truth only where the window covers a few tenths of a period (figure 7). The lag must be set from the oscillation period, which a constructed system supplies and a training log does not, so from the grokking logs we take changes and never a level (section 7).

6.3 Nuisance factors and nulls

At $r = 4$ we vary factors that leave r unchanged over five seeds and twelve observers, against the shift a real change of four components produces (table 9). A ramped observer gain moves the estimate by a component and a half, so a run whose parameter norm grows steadily will drift by that much at fixed r : the confound that matters most in section 7. Shifting the forcing band by an octave changes the roughness sharply while moving the estimate by about half as much.

Roughness and the estimate still move together there, so a control holds one fixed while the other changes. Two sums of four sinusoids are built with the same roughness to a part in 10^{15} , the first on a

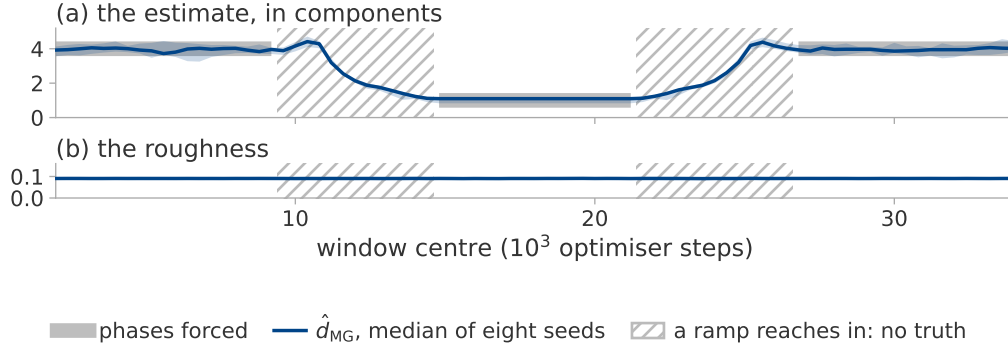


Figure 3: The estimate follows a change in the number of phases that leaves the roughness fixed. The harmonic control of Appendix M, scheduled from four phases to one and back; median over eight seeds, band their range. Grey, the number of phases, drawn only over windows holding one system; a window a ramp reaches into holds two and has no active dimension. The roughness stays between 0.085 and 0.094.

four-torus and the second, its frequencies harmonics of one irrational base, on a circle; a schedule passes between them and back. The estimate follows the number of phases and the roughness never moves (figure 3); under five invertible maps of the four-phase system the roughness rises by four fifths and the estimate by a tenth of a component (Appendix M).

The effective rank. Every construction in section 5 forces its r phases at equal amplitude, so the effective rank and the active dimension agree there and no experiment can say which of them the estimator reports. Forcing phase l at amplitude q^l keeps the path on an r -dimensional torus while the effective rank falls: the estimate stays closer to r , registering only the phases it can resolve (figure 8).

The reference band. The band drawn in section 7 is where the identifiability ratio fell on the accurate cases of figure 2; it records those cases and is not calibrated to hold beyond them.

7 Application to grokking

We apply the procedure to the two grokking settings of section 2: a transformer on mini-batches, which places it in the stochastic regime of section 3.3, and a full-batch perceptron (Appendix J lists every run).

7.1 The diagnostics reject both settings

We run the frozen configuration of section 5.1 on these logs, shortening the window and stride to fit the records and changing nothing else (table 12, figure 9). The regularised transformer runs are unstable in the embedding dimension, their estimates ranging from four to thirteen components across configurations. The perceptron runs and the transformer runs without weight decay carry the signature of a transient instead: two trend crossings in a whole record, at a ratio still far outside the band. Those undecayed runs are trained on mini-batches, so the regime label does not decide the outcome. The level is therefore not a dimension in either setting. A change within a run could still be one, since a fixed configuration adds the same offset to every window of that run; this window cannot show it, spanning tens of thousands of optimiser steps against a transition a few hundred steps wide.

Label-matched pairs. The perceptron targets come in pairs whose members fit the training set alike, only one of each representable by the architecture. The estimate separates them, but these runs are all transient and the raw loss curves separate them too (Appendices F and H.1).

7.2 The trajectory simplifies at generalisation

The effective rank is defined whether or not the path returns on itself, so a stored trajectory allows a measurement that a log does not. We store the trajectory of a transformer trained to grok, compressed

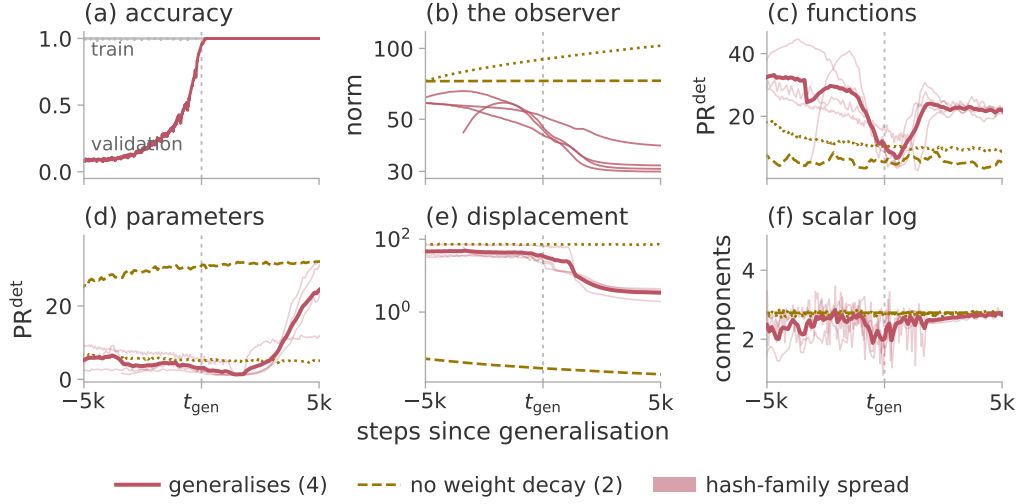


Figure 4: Simplification of the trajectory at generalisation. Three regularised seeds of modular addition and one S_5 composition against the same configurations without weight decay, each aligned on the generalisation step of the run it matches. Thick line, the mean of the four generalising runs; band, the hash families of Appendix G.1. (a) is the one run whose training curve is stored, (b) the observer for all six, (c)–(e) PR^{det} in each space and the displacement, over windows of 600 steps (300 for S_5), and (f) the estimate of section 7.3 on (b).

at every logged step by a random projection of the parameters and the probe logits (Appendix G.1). Near generalisation the detrended effective rank collapses in both spaces, reaching near unity in parameter space. In each space the minimum falls after the transition and within two thousand steps of it. Both then re-expand (figure 4, table 14). The collapse might follow training progress, but it occurs at generalisation in seeds whose generalisation steps differ by a factor of four and in a second task (figure 11). It might record a trajectory that has come to a halt, but the displacement reaches its floor thousands of steps later and never recovers. The effective rank measures variation inside the window, so only that variation collapses and $d_{\text{act}}(R)$ may be unchanged.

The runs without weight decay. The same two configurations trained without weight decay memorise and stay at chance accuracy. Within the transition window they decline much less than the generalising runs, but somewhere in training one of them declines just as far (table 14), so depth does not separate the generalising runs from the memorising ones. The timing of the decline does, as does whether it reverses.

7.3 The log estimate falls where the trajectory does

The test section 7.1 could not perform becomes possible once the window is set in units of the transition. A window that short need not lie inside one regime, so the fall below is a signal that tracks the transition and not a measurement of equation (1). We re-run the estimator on the same logs at the window length and midpoints the direct measurement uses, so both are sampled at the same instants. The frozen configuration does not fit a window that short, so we report every cell of a grid and take as headline the cell named by a rule blind to the outcome (Appendix H.3).

On the parameter norm the estimate halves at the generalisation step in all four generalising runs (figure 4f). That drop lies above the eighty-ninth percentile of the same statistic at every other window of each generalising run, while the other two show nothing remarkable there; all but two of the usable grid cells separate the generalising runs from the other two (table 16 and Appendix H.3). We also compare each run against randomised series built from it that keep its slow shape and its fluctuation spectrum while destroying its fine structure. The fraction of those falling as steeply as the run itself is smaller for every generalising run than for either of the other two (Appendix H.3).

The same event is thus recorded twice: once by storing the trajectory, once in a log a run writes anyway. The second record is the weaker, its observer the parameter norm, for which section 6.3

leaves a smoothness explanation open on a training log; the comparison against randomised series carries the claim alone, the roughness of these windows falling by a far larger factor than the estimate, which merely halves.

8 Conclusion

In a deterministic recurrent system with a matched delay lag, one scalar log of a suitable observer counts the active components of the regime, to within about one component and up to about eight. Outside that regime it does not: a transient supplies no returns; mini-batch noise contributes directions of its own. The estimator is usable only alongside the identifiability ratio and the trend-crossing count, which place both grokking settings outside that regime. Our result there comes from the stored trajectory, whose detrended effective rank collapses at generalisation and then re-expands. A memorising run can fall as deeply; only the timing of the fall and its reversal separate the generalising runs from the memorising ones.

Limitations. Every system in the evaluation is externally forced and training is not. The measurement needs a recurrent regime and a lag matched to a period no log reveals. Above about eight components it is only ordinal. We never check that the neighbour radii span a range over which the exponent is constant: they are whatever the twenty nearest neighbours give, and on a transient all twenty fall within thirteen per cent of one another. The collapse comes from four runs against two, distinguished by weight decay, which also drives the parameter norm; it marks the transition only once the transition has happened. It does not reproduce in the full-batch setting. The drop at the matched window is close to the drop a ramped gain alone produces, and a decline of the estimate is not by itself specific to generalisation (Appendix H.2). The modular run without weight decay has almost no range on that log, so its unremarkable value there records the observer and is not a negative result. On a circle the estimate is not geometric: two of eight seeds reproduce it from the base frequency alone (Appendix M). The edge-of-stability evidence comes from one task (Appendices B.3, H.1, H.3 and K).

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446 Appendix outline

447 These appendices are supplementary. The body stands without them; each holds the data, the
 448 construction or the audit behind a claim made there.

449 **Appendix A** states, without proofs, the four published results the estimator is assembled from: the
 450 delay embedding theorem of Takens [1981], its extension to fractal sets, the version for systems
 451 forced by an external input, and the maximum likelihood estimator of an intrinsic dimension. It
 452 closes with the limits a finite record imposes.

453 **Appendix B** records the procedure as run: the algorithm and its constants, the scalar observers
 454 of section 5.1, and the configurations at which the estimator was frozen, with the grids they were
 455 selected on.

456 **Appendix C** scores section 5 in full, observer by observer, against every simpler statistic we could
 457 compute on the same data. Three of those comparisons go against the estimator.

458 **Appendix D** gives the three experiments section 6 summarises, on the delay lag, the nuisance factors
 459 and unequal forcing amplitudes, and a fourth the body does not report: whether a real change in the
 460 active dimension is detected when one occurs.

461 **Appendix E** describes how the image-data system of section 5.1 is built in the three variants the paper
 462 scores, and measures how many independent directions each excites. The function-space variant falls
 463 short of its own requirement.

464 **Appendix F** gives the diagnostic values behind section 7.1 for every training log, and shows the
 465 label-matched pairs beside the raw loss curves that limit any claim of a separation between them.

466 **Appendix G** covers the measurement of section 7.2: the random projection that makes storing a
 467 whole trajectory affordable, the checks that validate it, the spectral guarantee those checks do not
 468 supply, and the collapse run by run.

469 **Appendix H** makes one argument three times: a statistic computed over a window cannot localise
 470 anything shorter than that window. The three parts make it for the full-batch trajectory, for a run
 471 whose estimate declines although it never generalises, and for the shortened window of section 7.3,
 472 where that result is also qualified.

473 **Appendix I** audits the Theiler exclusion, the one setting that fixes the value returned on a transient
 474 and the one at which this paper departs from its own protocol. It measures the cost of that departure
 475 and varies the exclusion from zero upwards.

476 **Appendix J** gives every run its task, architecture, length and outcome, and states where our grokking
 477 settings depart from the published configurations they are taken from.

478 **Appendix K** tests whether gradient descent at the edge of stability is deterministic and recurrent at
 479 once, as the estimate requires, and how much of that survives a coarser log.

480 **Appendix L** separates the level the estimator returns at high rank from the highest rank at which it
 481 still tracks the truth.

482 **Appendix M** varies the number of phases at a fixed roughness and the roughness at a fixed number of
 483 phases. It also reports two findings that count against the estimator.

484 **Appendix N** states the compute used and the material to be released.

A The results behind the estimator

The results below are stated without proof and in the form the paper uses. Every condition they impose is an assumption about the run rather than a property we can verify from the log.

Throughout, f is the map advancing the system by one step, ϕ the scalar observation, and

$$\Phi_{f,\phi}(s) = (\phi(s), \phi(f(s)), \dots, \phi(f^{E-1}(s))) \in \mathbb{R}^E \quad (4)$$

the delay map.

Theorem 1 (Delay embedding; [Takens, 1981](#)). *Let M be a compact manifold of dimension d and let $E > 2d$. For a generic pair (f, ϕ) with $f \in C^2(M, M)$ a diffeomorphism and $\phi \in C^2(M, \mathbb{R})$, the delay map equation (4) is an embedding of M into $\mathbb{R}^E$.*

An embedding is one-to-one and an immersion, so distances in $\mathbb{R}^E$ between reconstructed points are a faithful, though distorted, image of distances between states. The set they fill is the object the neighbour statistic below measures (figure 5). Theorem 1 requires a manifold and the sets a training run occupies need not be one; the extension below is the version the paper relies on.

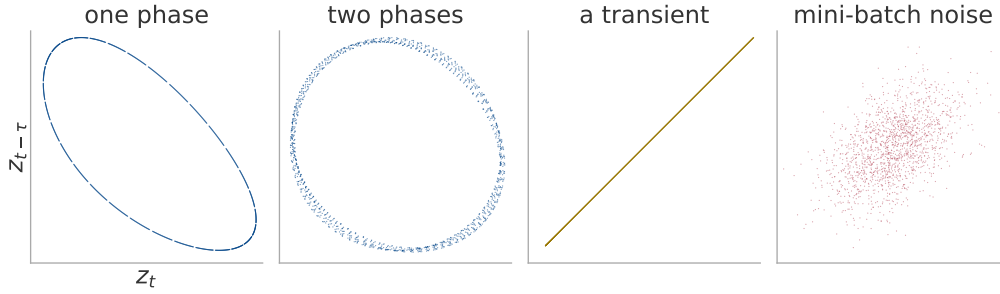


Figure 5: The set the neighbour search measures, in the plane of the first two delay coordinates. One window of the image-data system of section 5.1 under each excitation, reconstructed at the frozen configuration. A single phase closes an orbit, two fill a surface, a converging run never returns, and mini-batch noise fills the space it is given.

Theorem 2 (Fractal delay embedding prevalence; [Sauer et al., 1991](#)). *Let $A \subset \mathbb{R}^n$ be compact with box-counting dimension d , invariant under a smooth diffeomorphism f defined on an open neighbourhood of A , and let $E > 2d$ be an integer. Assume that for each $p \leq E$ the set of periodic points of f of period p has box-counting dimension less than $p/2$, and that the linearisation of f at each such orbit has distinct eigenvalues. Then for almost every smooth ϕ , in the sense of prevalence, the delay map equation (4) is one-to-one on A and an immersion on every compact subset of a smooth manifold contained in A .*

The dimension in the condition $E > 2d$ is the box-counting dimension of the occupied set $\mathcal{A}_R$, which need not be an integer, so the criterion applies to the sets a training run occupies and not only to tori. Prevalence, in turn, is a measure-theoretic notion of “almost every” on an infinite-dimensional space.

Mini-batch sampling makes the update depend on a draw from the data as well as on the state, so neither theorem above applies as stated.

Theorem 3 (Delay embedding for forced systems; [Stark, 1999](#), [Stark et al., 2003](#)). *Let $s_{t+1} = f(s_t, w_t)$ with the forcing w_t generated by a measure-preserving system on a space Ω , either deterministically or as an independent draw selecting among finitely many maps. Under genericity conditions on f and ϕ analogous to those of theorem 1, and for E exceeding twice the dimension of a fibre, the map sending a state to its delay vector equation (4) is an embedding of each fibre M_ω , that is, of the set of states compatible with one realisation ω of the forcing.*

The guarantee holds fibre by fibre, whereas the sample available in practice is one trajectory, which lies in a different fibre at every step. No invariant measure on a fibre is being sampled, so the neighbour statistic below estimates nothing.

Theorem 4 (Maximum likelihood intrinsic dimension; [Levina and Bickel, 2004](#), [MacKay and Ghahramani, 2005](#)). *Let points be drawn from a smooth density supported on a d -dimensional*

manifold embedded in $\mathbb{R}^E$. Model the points falling within a small ball about $\mathbf{y}_i$ as a homogeneous Poisson process in the d -dimensional radial variable, with the density constant over the ball. Writing $r_j^{(i)}$ for the distance from $\mathbf{y}_i$ to its j -th nearest neighbour, the maximum likelihood estimate of d from the first m neighbours of that point is

$$\hat{d}_m(\mathbf{y}_i)^{-1} = \frac{1}{m-1} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}. \quad (5)$$

Under this model the reciprocal is unbiased and the estimate itself is not.

The estimator is therefore a local statistic: it reports the exponent over the range of radii the m nearest neighbours span rather than the limit equation (1) takes. For $0 < \varepsilon_1 < \varepsilon_2$ that exponent at a point $\mathbf{c}$ is

$$d_R(\mathbf{c}; \varepsilon_1, \varepsilon_2) = \frac{\log \mu_R(B(\mathbf{c}, \varepsilon_2)) - \log \mu_R(B(\mathbf{c}, \varepsilon_1))}{\log \varepsilon_2 - \log \varepsilon_1}. \quad (6)$$

Its median over $\mathbf{c} \sim \mu_R$ is the finite-scale active dimension $d_{\text{act}}(R; \varepsilon_1, \varepsilon_2)$ of section 3.1, which equation (3) estimates by pooling over the N points of the record. It agrees with $d_{\text{act}}(R)$ when the same power of ε continues below ε_1 . A phase of smaller amplitude than ε_1 is invisible to it however long the record. A record of N points resolves a dimension of at most about $2 \log_{10} N$ under the argument of Eckmann and Ruelle [1992], a bound appendix L finds does not explain the ceiling we measure.

B The estimator, its configuration and the observers

B.1 The estimator

Algorithm 1 is the estimator as it was run, with the departures from equation (3) that affect the values reported.

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ ; embedding dimension E ; neighbours m ; Theiler exclusion W_T ; dither σ

```

1:  $z_t \leftarrow (x_t - \text{mean } x) / \text{std } x + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$  ▷ breaks exact ties
2:  $\mathbf{y}_t \leftarrow (z_t, z_{t-\tau}, \dots, z_{t-(E-1)\tau})$  for  $t = (E-1)\tau + 1, \dots, n$ 
3:  $S \leftarrow 0; N \leftarrow 0; n_{\text{dist}} \leftarrow 0; n_{\text{ratio}} \leftarrow 0$ 
4: for each  $\mathbf{y}_i$  do
5:    $r_1^{(i)} \leq \dots \leq r_m^{(i)} \leftarrow$  distances to the  $m$  nearest  $\mathbf{y}_j$  with  $|i-j| > W_T$ 
6:    $r_j^{(i)} \leftarrow \max(r_j^{(i)}, \varepsilon_{\text{dist}})$  for each  $j$ ; count those clipped into  $n_{\text{dist}}$ 
7:    $S_i \leftarrow \max(\sum_{j=1}^{m-1} \log(r_m^{(i)} / r_j^{(i)}), \varepsilon_{\text{ratio}})$ ; count clips into  $n_{\text{ratio}}$ 
8:    $S \leftarrow S + S_i; \quad N \leftarrow N + 1$ 
9: end for
10: mark the window degenerate if  $n_{\text{dist}}$  or  $n_{\text{ratio}}$  exceeds 1 % of its denominator
11: return  $(N(m-1) - 1) / S$ 
```

The returned value is $(N(m-1) - 1) / S$, the exactly unbiased pooled form, which differs from equation (3) negligibly at the sizes used here. A window marked degenerate is returned rather than discarded. No published cell of table 12 contains a marked window. The returned value is never clamped to E , so a divergent estimate is reported as divergent.

The Theiler exclusion is $W_T = \max((E-1)\tau, t_{\text{acf}})$, where t_{acf} is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples; appendix I.1 measures the effect of that cap. The floors $\varepsilon_{\text{dist}}, \varepsilon_{\text{ratio}}$ and the dither σ are $10^{-8}, 10^{-5}$ and 10^{-9} .

B.2 The observers

Table 1 defines the twelve observers of section 5.1.

Table 1: The scalar observers. $\mathbf{c}$ is the coordinate in the k -dimensional subspace and θ_0 the frozen head at initialisation; $\mathbf{L}$ is the matrix of probe logits, where the probe is a fixed held-out set drawn once before training; $\mathbf{g} = \partial\ell/\partial\mathbf{c}$. The projection directions $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{a}$, and the rotation $\mathbf{R}$, are drawn once from a fixed seed and never redrawn.

name	family	definition
instantaneous loss	loss	the loss on the current minibatch
full-batch loss	loss	the loss on the whole training set
probe loss	loss	cross-entropy on the probe set
parameter norm	norm	$(\ \theta_0\ ^2 + \ \mathbf{c}\ ^2)^{1/2}$
subspace norm	norm	$\ \mathbf{c}\ _2$
function-space norm	norm	$\ \mathbf{L}\ _F$
gradient norm	gradient	$\ \mathbf{g}\ _2$
gradient projection	gradient	$\langle \mathbf{u}, \mathbf{g} \rangle$
fixed parameter projection	projection	$\langle \mathbf{v}, \mathbf{R}\mathbf{c} \rangle$
function-space projection	function	$\langle \mathbf{a}, \text{vec } \mathbf{L} \rangle$
margin	function	mean probe logit gap, true class less best other
probe accuracy	function	accuracy on the probe set

546 B.3 Frozen configurations

547 One frozen configuration serves each range of the active dimension, each selected once on data
548 withheld from every later experiment (table 2).

Table 2: The frozen configurations and their selection. Only the swept parameters were varied; the remainder were fixed before any estimate was computed. Settings common to both configurations are not repeated: $m = 20$ neighbours, a Theiler exclusion of at least the embedding span (appendix B.1). The neighbour count and the exclusion were swept in the eight-component grid, in which the two Theiler rules return bit-identical values, so the selection between them is arbitrary.

	eight-component	twenty-component
$E_{\max}$	20	40
τ	4	16
window / stride	8 000 / 2 000	8 000 / 4 000
swept	$E_{\max}, \tau, m, \text{Theiler, window}$	$E_{\max}, \tau$
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–3	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.17	2.47
selection error, worst	2.03	3.35

549 The selection objective in each case is the error of the median estimate against the median measured
550 effective rank, over the four observers reserved for selection. The eight-component objective adds a
551 penalty on the rank correlation and on the spread across seeds.

552 *Each configuration lies at a grid boundary*, at the top of every axis it was swept over, so a wider grid
553 could move the level.

554 *The selection error spans a factor of twelve on identical data* (table 2), so the level is as much a
555 choice as a measurement. Only the withheld numbers of section 5.1 demonstrate recovery.

556 *The stride varies between experiments*. Table 3 states the window geometry each experiment uses;
557 none of them overrides a setting of the estimator. The constructed systems produced the recovery of
558 section 5.1 at a stride of their own, so “at the frozen configuration” is true there of the estimator and
559 not of the stride.

Table 3: The window and stride of each experiment, with their departures from the frozen configuration of table 2. Lengths are in samples of the log; the second stride applies to the longer records.

experiment	window	stride
frozen configuration	8 000	2 000
constructed systems (section 5)	8 000	3 000, or 3 666
training logs (section 7)	a third of the record	1 000

560 C Per-observer results and alternative statistics

Table 4: Recovery on each system, with the requirements of section 4 that each fails. Systems are ordered by increasing realism. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run’s sliding windows; ρ is over ranks after a median over seeds. The oscillating matrix has no optimiser, so the zero-learning-rate check does not apply to it. The function-subspace row fails constructed truth in half its withheld cells, which reach about $0.87r$ against the $0.9r$ the check asks.

system	truth	range	MAE	ρ	requirements failed
oscillating matrix	constructed	1–8	0.27	1.00	<i>not applicable</i>
online linear regression	constructed	1–20	0.79	0.99	frozen configuration, zero learning rate
logistic regression	constructed	1–20	0.81	1.00	frozen configuration, zero learning rate
frozen nonlinear decoder	constructed	1–20	1.58	0.94	zero learning rate
perceptron in a k -subspace	constructed	1–20	0.92	0.99	zero learning rate
image data, parameter subspace	measured	1–8	0.90	1.00	none
image data, parameter subspace	measured	1–20	1.75	1.00	withheld ranks
image data, function subspace	measured	1–8	0.28	1.00	all of them

561 All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld
562 seeds of section 5.1, in the deterministic recurrent regime, scored against the measured effective rank
563 rather than against r . Table 5 reconciles the three errors this paper quotes for that one measurement.

Table 5: Three errors for one measurement, and where each is quoted. Each row is the same eight-component recovery of section 5.1, on withheld ranks and withheld seeds in the deterministic recurrent regime, summarised in a different order. The order is stated wherever one of these figures appears.

aggregation	reported in	error
mean over (rank, seed, observer) runs	section 5.1, table 4	0.90
median over observers of each observer’s error	table 6	0.48
median over seeds and observers at each rank, then scored	table 8	0.382

Table 6: The accuracy of each scalar observer, against the smoothness that might explain it. One row per observer, except that the parameter norm and the subspace norm share one, their errors agreeing to two decimals. The slope is that of the estimate on the truth. The rank correlation is 1.00 for every observer, so it has no column. The last column records whether the roughness orders the withheld ranks perfectly.

observer	family	MAE	slope	roughness orders rank
probe loss	loss	0.60	0.82	no
function-space norm	norm	0.39	0.96	no
margin	function	0.46	0.83	no
parameter, subspace norm	norm	0.41	1.04	no
fixed parameter projection	projection	0.34	0.83	no
gradient projection	gradient	0.70	0.72	no
function-space projection	function	0.51	0.78	no
gradient norm	gradient	1.81	1.58	no
full-batch loss	loss	1.87	1.51	no
instantaneous loss	loss	0.61	0.73	no
probe accuracy	function	degenerate in every window		

Two observers are excluded from every aggregate. Probe accuracy is quantised, which collapses the delay vectors onto one another. The instantaneous mini-batch loss fails the zero-learning-rate check: frozen, it still climbs by more than five components across the withheld ranks and returns the estimate it returns when trained. The other eleven observers are exactly constant at zero learning rate, so no estimate is defined for them.

Nothing in a training log says which observer is the accurate one, so a reader can act only on the typical observer and not on the best row of table 6.

The highest rank correlation the roughness reaches over the four withheld ranks is 0.8, so smoothness alone accounts for no observer’s agreement with the truth (figure 6).

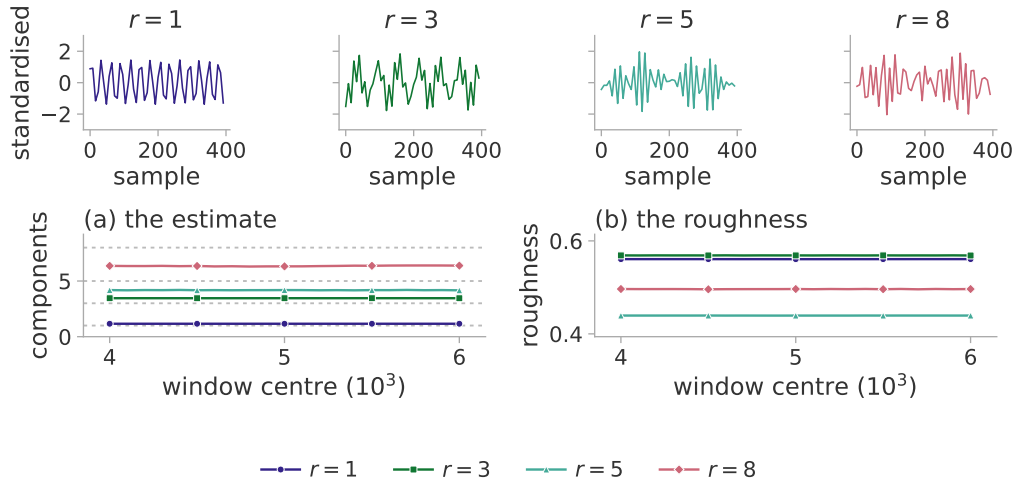


Figure 6: Four logs that look alike, separated by the estimator. The image-data system of section 5.1 at four ranks and a seed the configuration was not selected on, through the parameter norm. Above, four hundred samples of each log. (a) The estimate over sliding windows, the measured effective rank dotted. (b) The roughness of the same windows.

Table 7: At $k = 20$ the rank is tracked to about ten and then lost. Image data, median over three seeds and over the eight observers recorded there, which reports the typical observer rather than the best one; the truth is the measured effective rank. The last row is the spread of MG across seeds at fixed rank. Exploratory: this grid was scored with a smaller Theiler exclusion than Algorithm 1 asks (appendix I.1).

true r	1	2	4	6	8	10	12	14	17	20
MG	1.10	2.05	4.51	6.51	7.45	9.92	10.23	11.40	13.90	15.44
PR _{delay}	2.00	2.26	3.79	5.42	7.90	9.45	9.43	11.25	11.43	13.42
spectral PR	1.09	1.27	2.13	2.89	4.10	4.60	5.51	6.22	6.81	7.48
spread across seeds	0.87	0.51	1.35	1.81	2.64	3.59	2.81	2.07	3.41	4.24

Table 8: Every simpler statistic we could compute on the same data, scored beside the estimator on the image-data system of section 5.1. The two ranges are not aggregated alike: at $r \leq 8$ each statistic is scored on the withheld ranks alone, at $r \leq 20$ on all twenty, the selection ranks included, so only the comparison down a column is out of sample. Statistics marked $\dagger$ need a neighbour search; the rest are one-line computations on the series or its spectrum.

statistic	$r \leq 8$		$r \leq 20$	
	MAE	ρ	MAE	ρ
MG $\dagger$	0.382	1.00	1.75	0.997
LB $\dagger$	0.40	1.00	1.52	0.997
TwoNN $\dagger$	2.13	0.80	2.79	0.880
PR _{delay}	0.91	1.00	2.48	0.985
spectral PR, 256 bands	1.25	1.00	5.83	0.998
spectral PR, 1024 bands	1.08	1.00	4.78	0.979
spectral PR, native resolution	0.69	1.00	2.95	0.976
roughness	3.65	0.20	9.89	-0.379

Three qualifications attach to table 8. LB, the uncorrected per-point average, is more accurate than MG at $r \leq 20$: the correction of section 3.2 removes a positive bias, which on a saturating estimate was partly cancelling the saturation. The spectral participation ratio at 256 bands orders the ranks at $r \leq 20$ more faithfully than MG, at three times the error. The linear delay participation ratio needs no neighbour search at all, yet MG beats it by only a factor of 2.4 at $r \leq 8$.

D Conditions of validity in detail

The delay lag. Figure 7 varies the lag τ alone. Below a tenth of a period every rank above one returns about 2.5 and the ordering is lost; between three tenths and a half of a period the estimate follows the rank; above one period it diverges.

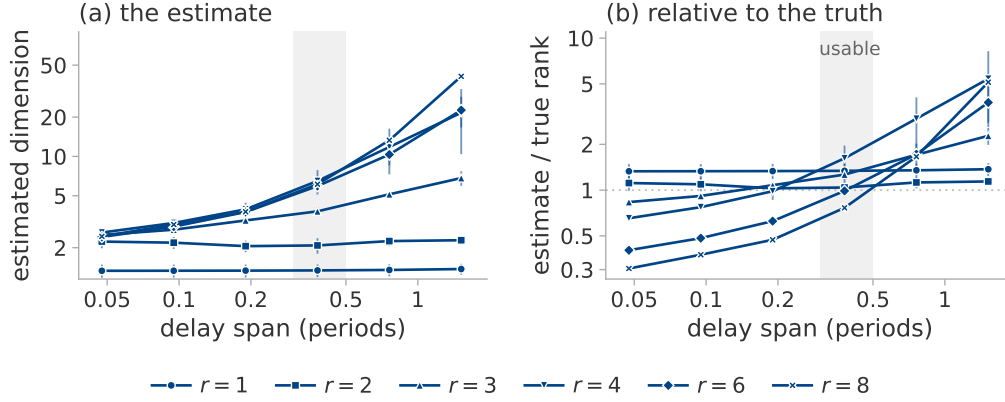


Figure 7: The delay lag, varied on one system: a fixed torus, two seeds, six lags spanning a twentieth of a period to one and a half periods. Marker shape is the true rank, the marker the median of the seeds and the bar their range. (a) The estimate. (b) The same divided by the true rank, so one reference line replaces six; the shaded column is three tenths to a half of a period.

582 **Nuisance factors.** The rotated-coordinate row of table 9 is scored on the fixed parameter projection
583 alone, since the other eleven observers are invariant to a rotation by construction; it measures a
584 property of the observer rather than of the estimator.

Table 9: Shift in the estimate produced by factors that leave the active dimension unchanged, median over five seeds and twelve observers. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale for the others. The last column is a percentage of a real four-component change, which the change-detection experiment below observes as a fall of 3.96. A constant rescaling of the observer moves the estimate by 0.000 and is omitted.

factor	$ \Delta MG $	as % of a real four-component change
none (baseline)	0.39	10
learning rate halved	0.37	9
forcing band up one octave	0.50	13
forcing band down one octave	0.78	20
coordinates rotated	0.28	7
observer gain ramped 10×	1.53	38
state amplitude ramped 4×	1.57	39

585 **Detection of a change.** We run three consecutive segments $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$, re-measuring
586 the ground truth on each. Over the three schedules and twelve observers, a deterministic recurrent fall
587 of 4.00 components is observed as 3.96, is detected both downward and upward on over nine tenths
588 of the runs, and recovers to within 0.05 of the level before it. Under stochastic excitation the same
589 true change is observed as 0.09. Table 10 gives each schedule separately, on the parameter norm.

590 The rule declares a change when the estimate falls below the median of the pre-switch segment by
591 three times the window-to-window scatter of that segment, for two consecutive windows.

Table 10: A change of the active dimension is recovered at both levels and after it reverses. Three schedules $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$, on the parameter norm. At $6 \rightarrow 2 \rightarrow 6$ the high level is over-reported by a component.

schedule	high	low	high again
$4 \rightarrow 1 \rightarrow 4$	4.06	0.84	4.06
$6 \rightarrow 2 \rightarrow 6$	7.00	2.17	6.95
$8 \rightarrow 3 \rightarrow 8$	8.32	3.34	8.28

592 **Unequal forcing amplitudes.** Figure 8 gives the measurement of section 6.3 in full, observed
 593 through the squared Frobenius norm and estimated with the frozen eight-component configuration.

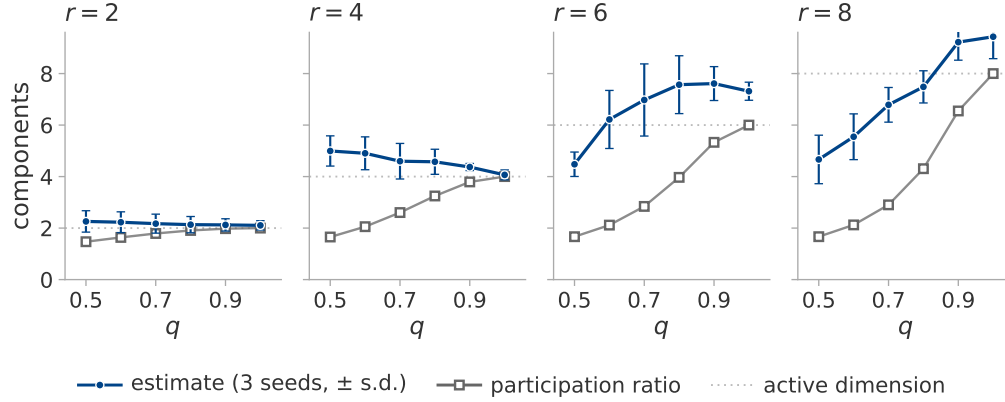


Figure 8: The estimate follows the active dimension, not the effective rank. Forcing phase l at amplitude q^l holds the active dimension at r while the effective rank falls with q . The system of section 5, three seeds, one panel per r ; the dotted line is r . Filled circles, the estimate, median over seeds, with standard deviation as bars; open squares, the measured effective rank, which matches its closed form to two decimals and so has no spread.

594 The estimate is flat in q at $r = 2$ and $r = 4$ while the effective rank declines, as intended, and declines
 595 itself at $r = 6$ and $r = 8$. The amplitudes show why: at $q = 0.5$ and $r = 8$ the weakest phase is
 596 forced at 0.008 of the strongest, far below the neighbourhood scale of the estimator, so the torus
 597 it resolves really does have fewer dimensions than the one that was built. The estimator therefore
 598 reports the number of phases it can resolve. That is neither r nor the effective rank; the three coincide
 599 only when the amplitudes are equal.

600 E System constructions

601 **The image-data system of section 5.1.** The data is partitioned into twelve fixed groups, the
 602 same partition at every rank. The loss weights of r of those groups are modulated by r rationally
 603 independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j .
 604 These directions are neither orthogonal nor of equal effect, since random data groups have correlated
 605 gradients, so a mixing matrix $\text{pinv}(\Phi)Q$ with Q an orthonormal basis of $\text{range}(\Phi)$ orthogonalises
 606 them. The per-phase scalar response gain of the linearised update is then divided out.

607 Table 11 measures how many independent directions each construction excites. The transient and
 608 the plain mini-batch rows report an outcome rather than a setting: a converging trajectory is a curve
 609 whatever r is and the rank of mini-batch noise is fixed by the per-example gradient covariance of the
 610 data, so projecting it onto r directions recovers only part of the intended rank and the estimates in
 611 that regime are scored against the measured value.

Table 11: The independent directions each construction excites, against the number intended. Measured trajectory participation ratio by excitation regime and nominal rank, median over the four seeds, for the ten available directions of section 5.1. Every estimate on this system is scored against these values, never against r . On the qp rows that carry the recovery claim of section 5.1 the two agree to three decimals and the choice moves no error by more than 0.001; it matters on the rows where the construction fell short.

regime	$r = 1$	2	4	6	8	excitation
qp	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast forcing
qp_slow	1.00	2.00	4.00	5.99	7.98	the same at a training-log timescale
qp_nopre	1.00	2.00	3.73	5.98	7.77	qp without the preconditioner
mixed	1.00	2.00	4.00	6.00	7.99	qp plus noise in the same directions
noise	1.00	2.00	4.00	5.99	7.99	injected rank- r gradient noise
noise_nopre	1.09	1.98	3.71	5.24	6.75	the same without the preconditioner
batch_proj	1.00	1.74	3.08	4.19	5.38	real mini-batch noise projected to rank r
batch	6.58	6.58	6.56	6.60	6.56	plain mini-batch descent
gd	1.04	1.05	1.04	1.04	1.06	a decaying transient
qp, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	the zero-learning-rate check

The functional rank of the backbone is 10 at every rank, seed and regime, a property of the frozen backbone rather than of the excitation.

The function-space construction. A local adapter is built around a trained network of widths 64, 16 and 10. Its function-space Jacobian is QR-whitened, so that the first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it. The functional, trajectory-covariance and update-covariance ranks are measured rather than assumed. Eight of the sixteen withheld cells fall short of the $0.9r$ effective rank the construction requires, at $0.87r$ in the median. Table 4 therefore marks this variant as failing every requirement.

The system at twenty available directions. The same construction at $k = 20$ reproduces the nominal rank over $r = 1, \dots, 20$ in the recurrent regime; every other regime behaves as it does at $k = 10$.

F Diagnostics and figures for the grokking application

Table 12 gives the diagnostics behind section 7.1.

Table 12: The estimate is a dimension in neither grokking setting. The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Every cell is a median over the run’s sliding windows. Seven of the seventeen logs figure 9 plots are shown here. A “no” is a run that had not generalised when training ended, which is a censored observation.

run	generalises	MG	ρ_{ident}	roughness	PR_{delay}	crossings
transformer, full data	yes	13.32	1.53	0.097	1.03	204
transformer, reduced data, s0	no	4.70	1.59	0.067	1.50	161
transformer, reduced data, s2	no	4.42	1.55	0.139	4.04	153
transformer, no weight decay	no	15.60	1.76	< 0.001	1.00	2
perceptron, $n + m$	yes	15.08	1.42	0.001	1.00	2
perceptron, $n \cdot m$	yes	14.97	1.39	0.001	1.00	2
perceptron, $n^3 + nm^2 + m$	no	15.41	1.51	< 0.001	1.00	2

Figure 9 places every training log of section 7 in the plane of the identifiability ratio and the trend-crossing count, against the region in which every accurate case of section 6 fell. No run lies in that region: every one is unstable in the embedding dimension, at ratios well above the 1.10 the region

allows. On the other axis the transformer and perceptron runs fail in opposite directions, since the perceptron series are monotone.

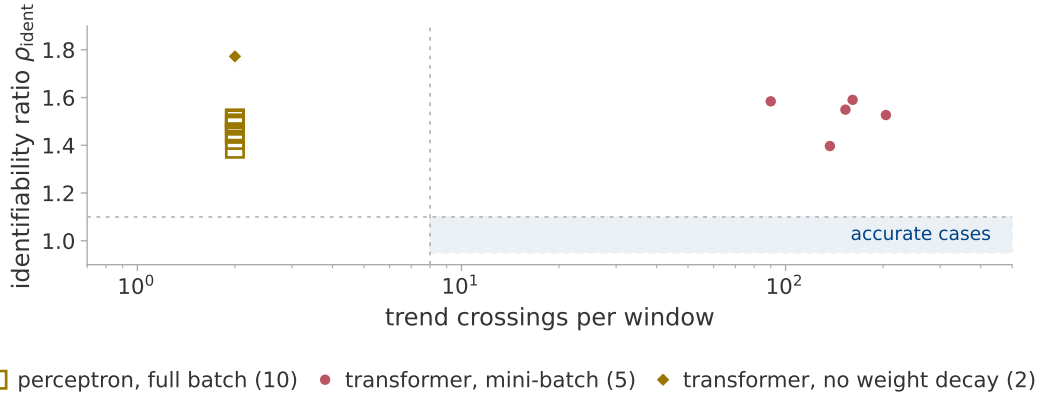


Figure 9: No grokking log lies where the estimate was accurate. Every log of section 7 in the plane of the identifiability ratio and the trend-crossing count, the transformer runs observed through the parameter norm and the perceptron runs through the training loss. Colour is the regime of section 3.3, marker shape the setting. The shaded rectangle is where every accurate case of section 6 fell. It is a record of those cases, not a test.

Figure 10 shows the separation of section 7.1 and the mechanism behind it: the medians separate every pair, though the ranges overlap.

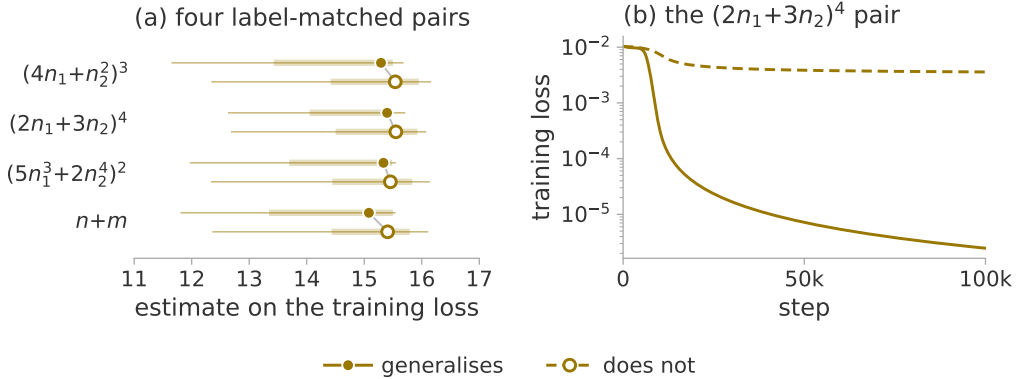


Figure 10: The estimate tells apart two targets that fit the training set alike, only one of them representable by the architecture. (a) The four pairs of section 7.1, on the training loss. Marker, median over the sliding window; thick bar, interquartile range; thin line, full range. All runs are transient, hence one colour; fill and dash mark generalisation. (b) One pair in full: the representable target drives the loss orders of magnitude lower.

G The direct trajectory measurement

G.1 The trajectory sketch

The direct measurement of section 7.2 needs the trajectory itself: the whole parameter vector at each of two to three thousand logged steps. Every logged step is therefore compressed to $\mathbb{R}^{1024}$ by a CountSketch [Charikar et al., 2002, Weinberger et al., 2009], a sparse random projection drawn twice from two independent hash families so that the two copies can be compared. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length, without which a signal taken from the logits could not be told from a measurement of the weight decay its scale is coupled to.

641 We do not verify a spectral guarantee for the compression. The two hash families supply a direct
642 estimate of the error instead: every reported value is the mean of the two projections, with the
643 disagreement between them recorded beside it. That disagreement is a per cent of the value at the
644 median and under nine per cent at its worst, smaller than every effect section 7.2 reports.

645 *The sketch is accurate on synthetic trajectories of known rank.* It agrees to within 0.11 components
646 throughout (table 13). Nothing above rank ten is tested, however, whereas the plateaus in figure 4 are
647 higher.

648 *It agrees with exact arithmetic on a real run.* On the perceptron architecture the participation ratio
649 computed on all the raw parameters agrees with the sketched value to three decimals.

650 *Recording the trajectory does not perturb training.* The log is required to be bit-identical with and
651 without the recording, over four hundred steps at reduced probe size. One global random stream
652 supplies the split, the initialisation and the mini-batch order in turn, so a single stray draw changes
653 the initial weights and can destroy grokking.

Table 13: The cost of the compression. The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in 145 500 dimensions. The difference between the first two columns is the cost of a sixty-sample participation ratio and is not attributable to the sketch.

true rank	PR, uncompressed	PR, sketched	sketch — uncompressed
1	1.00	1.00	0.00
2	1.99	1.99	0.00
5	4.56	4.54	−0.02
10	8.44	8.33	−0.11

654 Windows are sixty logged rows, each labelled by its centre, so a feature at a given step is located to
655 within half a window. The statistic reported per window is PR^{det} , the detrended effective rank of
656 section 3.1. The total displacement is recorded beside it, since a rank can fall simply because the
657 trajectory has stopped moving.

658 G.2 The effective-rank collapse, run by run

659 The plateau of table 14 is the median over the 4 000 steps before the generalisation step, the dip the
660 minimum within $\pm 4 000$ steps of it, and “at” the position of that minimum relative to it. The two runs
661 without weight decay have no generalisation step, so each is measured over the window of the run it
662 matches. In parameter space the post-dip maximum exceeds the plateau by an order of magnitude in
663 the modular runs. In function space it returns to about the plateau and exceeds it in two of the four
664 runs.

665 The same measurement rules out a collapse at one fixed absolute step. The four generalisation steps
666 span 9 980 steps and the minimum is searched over 8 000, so no single step lies inside the search
667 window of all four runs; every minimum in fact falls within 1 513 steps of its own run’s transition, in
668 both spaces (figure 11).

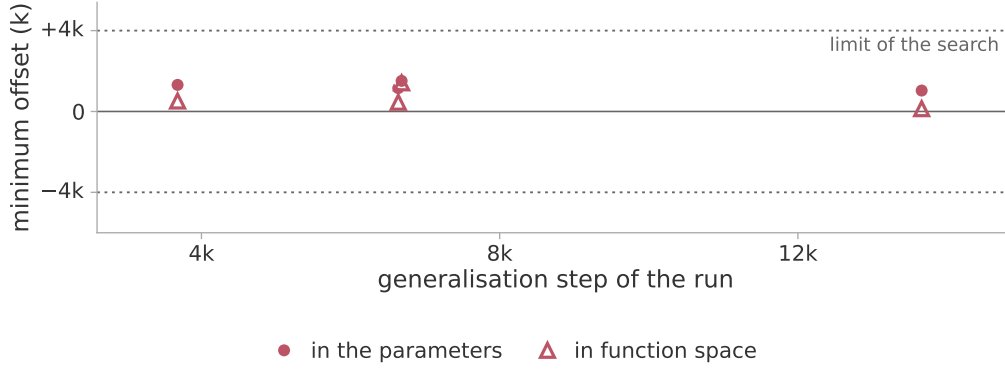


Figure 11: The collapse follows each run’s own transition. The minimum of the detrended participation ratio in each of the four generalising runs, against that run’s generalisation step. The two S_5 and modular runs near 6.7k nearly coincide. Dotted, the $\pm 4\,000$ steps searched around each transition.

Table 14: Plateau, dip and recovery per run, in function space and in parameter space. Depth is plateau divided by dip. The post-dip maximum is the largest value reached afterwards: the sense of “re-expands” in section 7.2. The two rows without weight decay in each block are measured in the aligned window; their deepest fall anywhere in their own run is 3.00 and 7.03 in function space and 1.66 and 2.39 in parameter space, the comparison section 7.2 also reports.

space	run	plateau	dip	depth	at	post-dip max
function	mod_wd1	24.24	3.80	6.38	135	23.99
	mod_wd1_s43	33.02	3.86	8.55	455	27.80
	mod_wd1_s44	27.16	3.40	7.99	515	30.82
	s5_wd1	21.04	2.62	8.03	1 413	23.53
	mod_wd0	5.50	3.43	1.60	—	—
	s5_wd0	11.87	9.09	1.31	—	—
parameter	mod_wd1	3.01	1.09	2.76	1 035	36.12
	mod_wd1_s43	2.57	1.15	2.24	1 155	37.34
	mod_wd1_s44	2.71	1.13	2.39	1 315	37.44
	s5_wd1	7.14	1.39	5.14	1 513	12.46
	mod_wd0	29.93	27.70	1.08	—	—
	s5_wd0	5.57	4.49	1.24	—	—

669 H The resolution of a windowed statistic

670 H.1 The full-batch setting

671 Repeating the direct measurement on the perceptron setting gives a limit of the measurement rather
672 than a negative finding. Under full-batch gradient descent every window of every run has a participa-
673 tion ratio between 1.00 and 1.36 on the statistics the comparison uses.

674 A deterministic update makes the trajectory a smooth curve. Over a 600-step window such a curve is
675 straight to within the measurement precision, so the statistic reports the curvature of the trajectory at
676 the scale the window sets, where there is almost none. The statistic has no range in which a collapse
677 could appear.

678 A range of window lengths settles the question (table 15). Two runs of 150 000 steps, a generalising
679 one and its label-matched partner, are measured over windows spanning 600 to 118 000 optimiser
680 steps, each holding the same sixty samples spread further apart as the window lengthens, so that
681 the bound on the statistic and its noise floor stay where the published measurement put them. The
682 statistic does leave its floor, at a scale of about 10^4 steps, more than an order of magnitude above the
683 published window. The absence of a feature at the shorter window was therefore a limit of resolution
684 rather than a fact about the run.

The feature that appears there separates nothing. One localised maximum appears, of the same size in both runs (table 15); function space agrees, with more range and no more separation. Nor can a window this long place that maximum to better than the interval between memorisation and generalisation. Beyond 30 000 steps the ordering even reverses and the non-generalising run moves further, because the window is then a large fraction of a run whose weight norm is still climbing, so that the statistic tracks that drift. The full-batch trajectory therefore carries no signature of generalisation in its effective rank at any window length between 600 and 118 000 steps: the collapse of section 7.2 belongs to the regularised stochastic setting.

Table 15: The full-batch statistic leaves its floor only at the longest windows. Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window, sixty samples per window at every length.

window (steps)	generalises		control, never generalises	
	median	max	median	max
600	1.002	1.035	1.001	1.011
2 950	1.066	1.258	1.003	1.087
5 900	1.027	1.534	1.005	1.360
11 800	1.010	1.951	1.020	1.826
29 500	1.006	2.047	1.112	1.735
59 000	1.025	1.551	1.345	1.912
118 000	1.231	1.626	2.009	2.271

Introducing mini-batches confirms this. The same statistic then ranges over more than a decade while the generalisation step moves by a single logged step. Any claim based on a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

H.2 Declines of the estimate that are not generalisation

A decline of the estimate is not specific to generalisation. In a long run at $p = 211$ with no weight decay, which memorises early and ends at chance validation accuracy, a decline is detected tens of thousands of steps after memorisation. The ramped-gain experiment of section 6.3 supplies a mechanism for it that needs no dimension at all: the parameter norm of this run roughly triples, acting on the estimate through the observer. No trajectory was stored for that run, so nothing here says whether it simplified; at the frozen window and in these settings the estimate on a log is in any case neither a dimension nor an effective rank.

Section 7.3 does not make the decline specific either. The estimate on a log does decline where the trajectory does, in four runs and in neither of the other two, but at a window some seventy times shorter than the one used here, under a configuration chosen for that window.

H.3 Resolution of a windowed estimate on a training log

The change section 7.1 leaves open is testable on the seven extended runs. The frozen configuration takes a window of a third of each record, advanced by 10 000 steps at a time. That gives nine windows per run and localises a feature only to $\pm 20\,000$ steps. The runs therefore cannot be aligned on their generalisation steps as figure 4 aligns them: no two share an offset, so no mean over runs can be formed.

The one run with a window centre near its generalisation step *rises* by 2.93 components over the following stride and by 2.71 over the one after. Those are the only two of the fifty-six stride-to-stride changes to exceed the 1.53-component floor a ramped gain alone produces (section 6.3). The other two generalising runs stay inside it throughout, as do all four that never generalise (figure 13). A rise is not the fall section 7.3 reports, and at $\pm 19\,995$ steps neither could be assigned to the transition.

As in appendix H.1, the statistic has no range in which a feature confined to the transition could appear. Testing the claim requires a window set in units of the width of the transition, which for these runs is hundreds of steps.

721 **The matched-window re-run.** Section 7.3 performs that re-run on the six runs of figure 4 rather
722 than the seven of this appendix: the question needs a paired comparison and only those six had their
723 trajectory stored. Each window is 60 samples, centred on the midpoint of a window of section 7.2.
724 The significance of the fall is assessed against the surrogates of section 7.3 [Theiler et al., 1992,
725 Schreiber and Schmitz, 1996].

726 *The frozen configuration does not fit.* At 60 samples its delay span of 76 exceeds the window. The
727 grid crosses $E_{\max} \in \{4, 6, 10\}$, $\tau \in \{1, 2, 4\}$, $m \in \{5, 20\}$ and both Theiler rules, giving thirty-six
728 cells, of which twenty-six are long enough to return a value. Twenty-four of those separate the four
729 generalising runs from the other two. The headline cell is named by the rule of section 7.3, which
730 does not choose the best cell: a neighbouring cell separates the runs more sharply.

731 *The diagnostics do not survive the shortening.* The identifiability ratio needs an embedding of $2E_{\max}$,
732 so at the headline cell the Theiler exclusion leaves too few usable points to compute one. Where the
733 ratio can be computed at all, the two Theiler rules disagree about whether it passes; it settles nothing.

734 *Linear detrending removes the effect.* Repeating every cell on a linearly detrended window leaves
735 none that separates the generalising runs from the rest. Detrending a sketch and detrending a scalar
736 are not comparable: on a 1024-dimensional sketch a per-coordinate linear detrend removes one
737 direction out of many, whereas on a scalar series it removes most of the series.

738 *The observer matters.* On the training loss rather than the parameter norm, twelve of the twenty-six
739 cells separate them, though the surrogate comparison passes in the S_5 run only.

740 *The two runs without weight decay have almost no dynamic range.* At this window the baseline
741 belongs to the configuration rather than to the run: on a standardised straight line equation (3) returns
742 about 2.77, close to the pre-transition level of every run. The parameter-norm log of the modular
743 run is straight to under a per cent of its own spread, so its estimate spans three thousandths of a
744 component against a range three orders of magnitude larger in the generalising runs; the S_5 run
745 is a weaker case of the same thing. Their D and quantile columns therefore record an observer
746 that offered the estimator no local structure to respond to; neither run is a negative result. The four
747 generalising runs are untouched, and so is the surrogate comparison, in which each run is measured
748 only against surrogates of itself (figure 12).

Table 16: The fall of the log estimate at generalisation. The matched-window re-run of section 7.3 on the parameter-norm log, at the headline cell $E_{\max} = 10$, $\tau = 1$, $m = 20$, embedding-span Theiler. D is the median estimate over the three thousand steps before the transition divided by the minimum over the window that brackets it; the quantile is the rank of D among the same statistic at every window centre after memorisation; the floor is where that minimum falls, relative to t_{gen} . The p columns are the fraction of surrogates, at three smoothing lengths, whose D is at least the observed one.

run	D	quantile	floor	p_{101}	p_{201}	p_{401}
modular, seed 42	2.96	0.99	−65	0.025	0.025	0.025
modular, seed 43	1.98	0.93	255	0.050	0.050	0.025
modular, seed 44	2.14	0.92	315	0.050	0.075	0.025
S_5 composition	2.12	0.89	1 313	0.050	0.050	0.050
modular, no decay	1.00	0.46	−865	0.950	0.950	0.925
S_5 , no decay	1.04	0.38	1 613	0.975	0.975	1.000

749 Put through the same comparison, the linear participation ratio responds less than the estimator and
750 the roughness, as section 7.3 reports, far more.

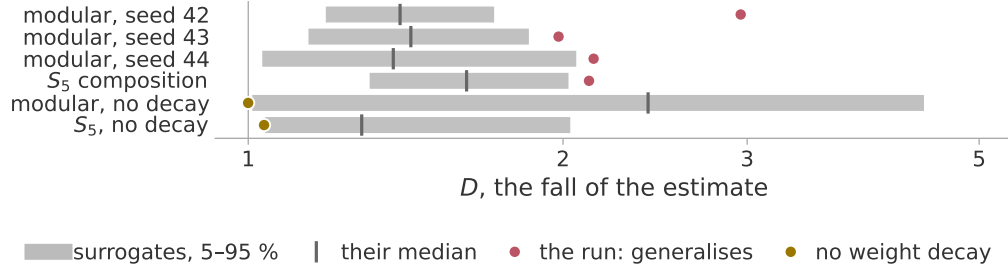


Figure 12: Each run’s fall against the null distribution built from that run alone. D is as in table 16, on the parameter-norm log at the headline cell. Band, the fifth to ninety-fifth percentile of the pooled surrogate draws, thirty-nine at each of five seeds; tick, their median; circle, the run. The runs without weight decay are measured in the window of the run each matches.

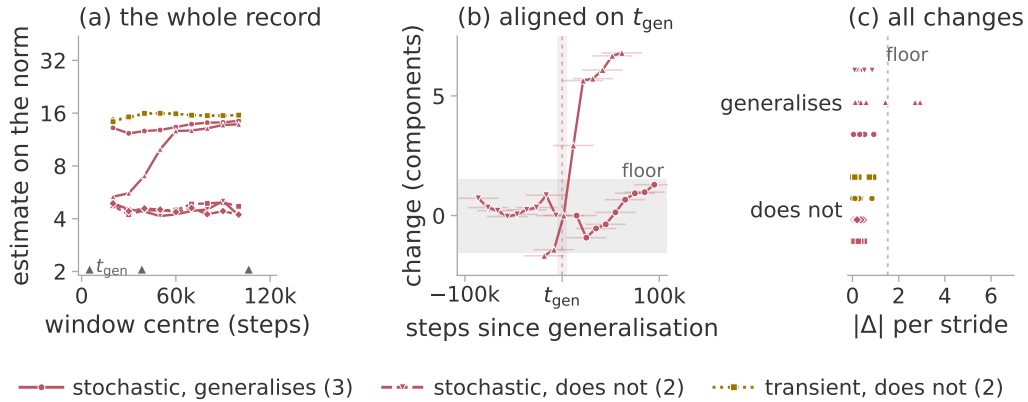


Figure 13: The frozen window localises a feature only to $\pm 20\,000$ steps. The estimate on the parameter norm for the seven extended runs, under the frozen configuration of section 7.1, plotted at the window centres. Colour is the regime of section 3.3. (a) Every window, with the generalisation steps marked on the axis. (b) The generalising runs aligned on their own generalisation step. (c) All the changes from one stride to the next, one sub-row per run. The grey band in (b) and (c) is the floor a ramped gain alone produces (section 6.3).

751 I The Theiler exclusion

752 I.1 The exclusion cap at the twenty-component configuration

753 The rule of appendix B.1 asks for at least the full span of a delay vector, so that two neighbours
 754 cannot be near each other merely because their windows overlap; the implementation caps that at 150
 755 samples.

756 At the eight-component configuration the span is 76, so the cap never binds and every result in
 757 sections 5 to 7 is scored at 76. At the twenty-component configuration the span is 624 and the cap
 758 binds unconditionally, which is why table 7 marks that result exploratory.

759 Table 17 measures the cost of that departure.

Table 17: The cost of the exclusion cap. The twenty-component system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured PR^{pos} and is not comparable with table 4, which covers more observers.

r	measured PR^{pos}	MG at $W_T = 150$	MG at $W_T = 624$
2	2.00	1.74	1.81
5	5.00	4.16	5.15
8	8.00	6.28	6.71
11	11.00	9.54	10.26
14	14.00	9.81	9.37
17	17.00	14.40	14.19
20	19.99	15.92	16.60
mean absolute error		2.16	1.89
Spearman ρ		1.00	0.96

760 The ordering of the ranks, all that section 5.1 claims at this range, is nearly unaffected, a single
761 inversion apart. The mean absolute error falls when the protocol is enforced, so the cap was not
762 making the reported result look better than it is. The saturation above about six components is present
763 at both settings and is therefore not an artefact of the exclusion.

764 The identifiability ratio is where the cap does bite. Its numerator is computed at $2E_{\text{max}}$ with τ held
765 fixed, so that span of 156 exceeds the cap while the denominator’s 76 does not. Every ratio in this
766 paper is therefore 76 against 150, the reference band of section 6.3 and the grokking logs alike, so
767 the comparison in section 7.1 is like for like; but it is not the fixed-exclusion quantity section 3.4
768 describes.

769 I.2 The Theiler exclusion and the dimension of a transient

770 Section 3.3 claims that recurrence is a requirement of this protocol and not of the estimator. The
771 claim is testable on the excitation cases of section 5.1 by varying the Theiler exclusion W_T alone,
772 from zero up to the value the frozen rule gives (table 18).

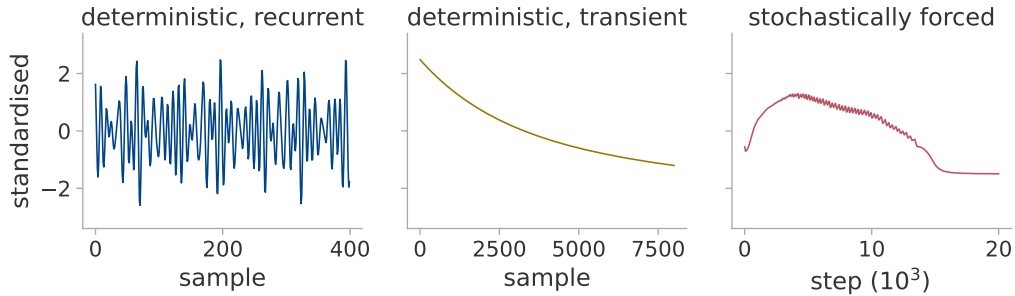


Figure 14: The three regimes as raw scalar logs. Each panel is the parameter norm, standardised to zero mean and unit variance as in Algorithm 1. Left, the constrained head of section 5.1 under fast quasiperiodic forcing, recurrent on a torus. Middle, the same system under plain gradient descent, transient over its whole record. Right, a transformer grokking modular addition, stochastically forced.

Table 18: The estimate against the Theiler exclusion, median over ranks, seeds, observers and windows. At $W_T = 0$ the transient returns the dimension of its own path, which is not the dimension of a regime (section 3.3); the exclusion removes the neighbours that value rests on. The fast recurrent case is unchanged under the same variation. Column 76 is the exclusion the frozen configuration asks for and the one every other result uses; the last removes the cap of 150 and applies the autocorrelation rule, some 1 700 samples here.

W_T	0	2	5	20	76	100	150	uncapped
recurrent, fast	4.83	4.82	4.82	4.82	4.80	4.80	4.80	4.80
recurrent, slow	2.06	3.02	3.22	3.27	3.27	3.27	3.28	3.27
transient	1.20	1.71	2.38	5.51	16.12	20.50	29.22	173.19

773 *The value at $W_T = 0$ is the dimension of the path.* On a uniformly sampled curve the nearest
774 neighbours of a point are its immediate temporal neighbours, for which equation (3) returns 1.199;
775 the transient returns that value in every cell. The excess over unity is the lattice bias of a deterministic
776 sampling.

777 *The exclusion removes the near neighbours, not the neighbours.* The estimator fills its twenty slots at
778 every exclusion; only the distances change. Before any exclusion the nearest neighbours of a point
779 lie thousands of samples away under fast quasiperiodic forcing, but six samples away on the transient
780 (figure 14). Excluding 150 samples therefore costs the fast forcing almost none of its close pairs and
781 the transient all of them (figure 15). The distances the transient retains are then nearly equal, their
782 log-ratio sum collapses, and the reciprocal in equation (3) diverges. An exponential decay with no
783 optimiser and no data behind it reproduces both ends.

784 *The asymmetry is general; the level is not.* Under fast forcing $W_T = 0$ already returns r , because that
785 case is not oversampled; it is the slow forcing that the exclusion repairs, its nearest neighbours lying
786 some seven hundred samples away.

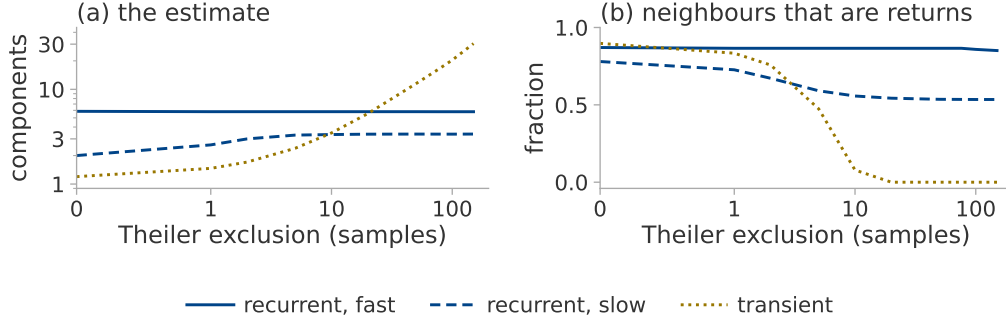


Figure 15: The exclusion costs a transient every close pair and a torus almost none. The excitations of table 18 on the parameter norm, median over ranks, seeds and windows. In (b) a neighbour counts as a return when it is no further away than the nearest neighbours an unexcluded search finds, so a fall to zero leaves the estimator nothing but distant points.

787 **A recurrence-based diagnostic.** The ratio $\hat{d}_{MG}(W_T=150)/\hat{d}_{MG}(W_T=20)$ is unity in both recur-
788 rent cases and above five on the transient, so it tests recurrence in the reconstructed space where the
789 trend-crossing count tests only non-monotonicity of the series: the gap section 3.4 leaves open, which
790 appendix K shows to matter on real runs. The principled version is older than either: a space-time
791 separation plot [Provenzale et al., 1992] shows directly how neighbour distance depends on separation
792 in time, and the exclusion should be chosen from one.

793 J The training runs

794 Table 19 lists every training run the paper draws on. **T** is the one-layer transformer of Nanda et al.
795 [2023]: width 128, four heads, no layer normalisation, 226 816 parameters on modular addition and
796 228 608 on S_5 , trained in double precision with AdamW. **P** is the two-layer quadratic perceptron of

797 [Gromov \[2023\]](#): width 500, 145 500 parameters at $p = 97$, trained by full-batch gradient descent
798 with no regularisation.

799 [Nanda et al. \[2023\]](#) train at full batch and we use mini-batches. In the perceptron setting we take the
800 tasks and the representability criterion of [Doshi et al. \[2024\]](#) but not their optimiser, running plain
801 full-batch gradient descent on mean squared error instead. The S_5 task [\[Power et al., 2022\]](#) and its
802 weight decay lie outside the configuration of [Nanda et al. \[2023\]](#). The two S_5 runs also step the
803 optimiser twice per batch, so their effective learning rate and decay are about double the nominal
804 ones.

805 **Definition of memorisation and generalisation.** The memorisation step is the first logged step at
806 which the training accuracy reaches 95 %; the generalisation step is the first at which the validation
807 accuracy does. Neither requires the threshold to be held thereafter. The extended reruns of section 7.1
808 use the stricter rule instead, requiring the threshold to hold over a sustained block.

Table 19: Training runs used in this paper. α is the training fraction, “final” the validation accuracy at the end of training. A dash in the generalisation column marks a run that had not generalised when training ended; such runs are censored observations, not established negatives.

run	task	arch	wd	α	steps	mem.	gen.	final	used in
<i>Trajectory sketch attached</i>									
mod_wd1	mod. add. $p=113$	T	1.0	0.30	20k	1470	13 660	1.00	section 7.2
mod_wd1_s43	mod. add. $p=113$	T	1.0	0.30	20k	1580	6640	1.00	section 7.2
mod_wd1_s44	mod. add. $p=113$	T	1.0	0.30	20k	1410	3680	1.00	section 7.2
s5_wd1	S_5 composition	T	0.2	0.50	15k	1280	6685	1.00	section 7.2
mod_wd0	mod. add. $p=113$	T	0.0	0.30	20k	630	—	0.03	section 7.2
s5_wd0	S_5 composition	T	0.0	0.50	15k	1015	—	0.01	section 7.2
<i>Extended reruns</i>									
gropos_s0	mod. add. $p=113$	T	1.0	0.30	120k	1450	5070	1.00	section 7.1
lowdata15_s0	mod. add. $p=113$	T	1.0	0.15	120k	620	—	0.58	section 7.1
lowdata15_s1	mod. add. $p=113$	T	1.0	0.15	120k	630	106 380	1.00	section 7.1
lowdata15_s2	mod. add. $p=113$	T	1.0	0.15	120k	660	—	0.01	section 7.1
lowdata20_s0	mod. add. $p=113$	T	1.0	0.20	120k	800	38 380	1.00	section 7.1
wd0_s0	mod. add. $p=113$	T	0.0	0.30	120k	610	—	0.09	section 7.1
wd0_s1	mod. add. $p=113$	T	0.0	0.30	120k	630	—	0.07	section 7.1
p211_wd0	mod. add. $p=211$	T	0.0	0.16	200k	1800	—	0.04	section 7.1
<i>Perceptron, $p = 97$, full batch, no regularisation</i>									
a_add	$n + m$	P	0	0.5	100k	9230	11 760	1.00	section 7.1
a_mul	$n \cdot m$	P	0	0.5	100k	9100	11 560	1.00	section 7.1
a_sub	$n - m$	P	0	0.5	100k	9110	11 540	1.00	recorded only
a_sq_sum	$n^2 + m^2$	P	0	0.5	100k	8440	10 470	1.00	recorded only
a_sum_sq	$(n + m)^2$	P	0	0.5	46k	7350	8190	1.00	recorded only
x_mix_quad	mixed quadratic	P	0	0.5	100k	9600	—	0.01	figure 9
x_no_grok	$n^3 + nm^2 + m$	P	0	0.5	100k	9860	—	0.01	section 7.1
g_p1	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7470	12 290	1.00	section 7.1
g_p1x	$+ n_1 n_2$	P	0	0.5	100k	10 140	—	0.01	section 7.1
g_p2	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6470	7680	1.00	section 7.1
g_p2x	$- n_1^2$	P	0	0.5	100k	9730	—	0.02	section 7.1
g_p3	$(5n_1^3 + 2n_2^4)^2$	P	0	0.5	100k	5970	6600	1.00	section 7.1
g_p3x	$- n_2$	P	0	0.5	100k	9420	—	0.74	section 7.1

809 Four of the perceptron runs are repeated with the trajectory sketch attached, for the measurement
810 of appendix H.1: a_add and x_no_grok from the table above, together with the label-matched pair
811 g_p1 and g_p1x. Each is run at full batch and again with mini-batches of 512. The full-batch runs
812 reproduce the memorisation and generalisation steps of the rows above exactly and the mini-batch
813 runs to within six logged steps, so appendix H.1 compares two noise levels on matched learning.

814 g_p3x ends far above the majority-class baseline of 1.3 % while being provably outside the class of
815 functions the architecture can represent.

816 K Descent at the edge of stability

817 Descent at the edge of stability [Cohen et al., 2021] is gradient descent at a fixed rate: the top Hessian
 818 eigenvalue sharpens until it reaches $2/\eta$ and then stays there, while the loss descends over long
 819 stretches and not over short ones. Such a run is deterministic and is not a plain transient, so an
 820 unforced trajectory could satisfy the conditions of section 3.3.

821 **Setting.** We run full-batch descent without weight decay on the quadratic perceptron of appendix J
 822 (table 20). The log is written at *every* optimiser step, because the oscillation is the two-cycle of
 823 the unstable mode and its period is a few steps. The sharpness is measured by power iteration on
 824 Hessian-vector products, taking the largest *algebraic* eigenvalue, since the Hessian is indefinite early
 825 in training.

Table 20: Oscillation and recurrence in unforced descent at the edge of stability. Full-batch descent by learning rate, both seeds. The stability ratio is $\eta\lambda_{\max}/2$ over the second half of the run; *rises* is the fraction of post-transition steps on which the loss increases; *return* is the nearest neighbour surviving a 150-sample exclusion divided by the distance the trajectory travels during that exclusion, for which a two-frequency torus scores 0.008.

η	stability ratio	rises	return	outcome
1×10^5	0.239, 0.238	0.000	1.007	monotone
3×10^5	0.758, 0.745	0.000	1.007, 1.007	monotone
1×10^6	0.969, 0.971	0.495, 0.490	0.104, 0.079	at the edge
1.5×10^6	0.974, 0.970	0.500, 0.500	1.012, 1.011	at the edge
2×10^6	0.977, 0.978	0.500, 0.500	0.120, 1.013	at the edge
2.5×10^6	—, 0.988	—, 0.500	—, 0.034	one seed diverges
2.8×10^6	0.992, —	0.503, —	0.022, —	one seed diverges
3×10^6	—	—	—	diverges

826 **Oscillation is not recurrence.** Eight runs reach the edge and all eight oscillate, the loss rising on
 827 one step in two, but only five of them return close to states already passed through. The return column
 828 of table 20 separates those two groups by a factor of eight with nothing in between. The split is not a
 829 threshold: at $\eta = 2 \times 10^6$ one seed returns and the other does not, at 0.120 against 1.013, while their
 830 sharpness and their fraction of rising steps agree to three decimals.

831 **Performance of the diagnostics.** The band of section 6.3 rejects all eight runs at the edge, on the
 832 training loss and on the parameter norm alike, whether or not they recur, ρ_{ident} lying far outside it in
 833 both groups while the trend-crossing count, which tests non-monotonicity and not recurrence, passes
 834 every one of them. The estimate does not separate the runs that return from those that do not either:
 835 their ranges on the training loss overlap.

836 **Dependence on the logging stride.** Written at the stride every other full-batch log here uses, most
 837 of the phenomenon disappears (figure 16): over these eight runs the median trend-crossing count
 838 falls from 71 to 6. Whether a log looks like a transient is thus in part a property of how often it was
 839 written.

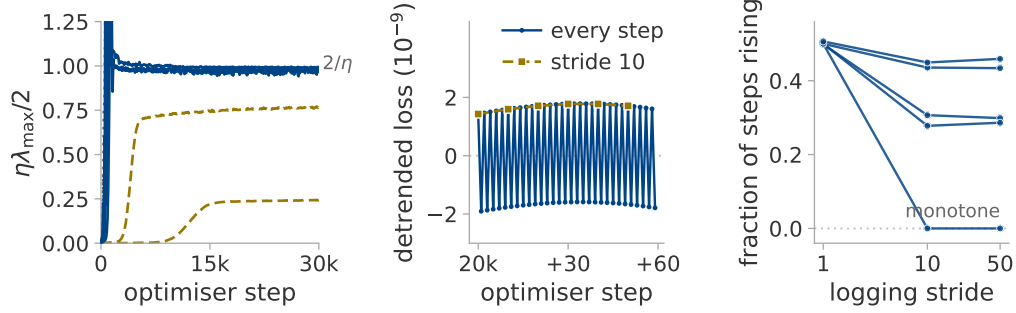


Figure 16: Full-batch descent reaches the edge of stability. (a) The stability ratio along training, one seed per rate. (b) Sixty consecutive steps at one rate at the edge, against the same series at a stride of ten: the two-cycle is removed by that stride rather than blurred by it. (c) The fraction of steps on which the loss rises, against the logging stride.

840 L The ceiling: embedding dimension against record length

841 Two explanations were available for the ceiling of section 5.1: the embedding condition $E > 2d$ and
 842 the finite-record bound $2 \log_{10} N$ [Eckmann and Ruelle, 1992]. Each is insensitive to the variable the
 843 other names, so we sweep $E_{\max}$ at a fixed record length and the record length at fixed $E_{\max}$, over
 844 $r \in \{2 \dots 20\}$ with everything else frozen (table 21). Each record length is a *prefix* of one longer
 845 record rather than a resampling of a fixed span, leaving the delay lag its meaning in periods. Every
 846 trajectory is bit-identical to the published one over their common prefix.

Table 21: The ceiling follows neither the embedding condition nor the finite-record bound. Two sweeps, three seeds per cell. *Tracking* is the largest r at which the median estimate stays within one component of the truth; *level* is the median estimate at $r = 20$; *slope* is $d\hat{d}_{MG}/dr$ over $r \geq 8$, which is 1 if the estimator still tracks the rank and 0 if it has stopped.

$E_{\max}$ at $N = 8000$	10	14	20	28	40	56	
tracking	6.5	7.3	8.1	10.3	10.9	11.2	
$E_{\max}/2$ predicts	5	7	10	14	20	28	
level at $r = 20$	6.9	8.1	9.8	11.3	13.2	14.9	
slope	0.08	0.12	0.20	0.29	0.41	0.54	
N at $E_{\max} = 20$	10^3	$2 \cdot 10^3$	$4 \cdot 10^3$	$8 \cdot 10^3$	$1.6 \cdot 10^4$	$3.2 \cdot 10^4$	$6.4 \cdot 10^4$
tracking	6.7	8.3	8.5	8.1	8.9	9.5	9.5
$2 \log_{10} N$ predicts	6.0	6.6	7.2	7.8	8.4	9.0	9.6
level at $r = 20$	9.0	9.3	9.4	9.8	9.9	10.2	10.5
slope	0.22	0.17	0.17	0.20	0.19	0.19	0.22

847 **Outcome.** The level at high rank is set by the embedding: a sixty-fourfold increase in the record
 848 adds 1.5 components, whereas a sixfold increase in $E_{\max}$ adds eight. The tracking ceiling follows
 849 neither variable. It never passes 11.4, yet at a quarter of the record the rest of the study uses it still
 850 tracks to 8.3, where $2 \log_{10} N$ allows only 6.6. A bound set by the record length cannot produce that,
 851 so Eckmann and Ruelle [1992] is not the explanation here. The embedding condition $E > 2d$ fails
 852 too in the form we used it: the ceiling rises by a tenth of a component per unit of $E_{\max}$ rather than by
 853 a half, so $E_{\max}/2$ happens to be right near $E_{\max} = 14$ and is wrong at both ends.

854 Fitting the level at high rank separates the candidates (table 22): neither hypothesis fits, with or
 855 without a free scale, and a form logarithmic in both variables absorbs the residual variation. That
 856 form has three parameters on sixteen cells and no held-out cell, so it describes the surface and does
 857 not explain it.

Table 22: Neither candidate explanation of the ceiling accounts for the measured level. Each form scored at $r \geq 16$ over the sixteen measured $(E_{\max}, N)$ cells, in sample: no cell is held out, so the error falls with the parameter count by construction. The spread of the fitted quantity itself is 2.00 components.

form	parameters	root-mean-square error
$E_{\max}/2$, the embedding condition	0	8.73
$2 \log_{10} N$, the finite-record bound	0	3.68
the pointwise minimum of the two	0	3.70
$0.56 E_{\max}/2$	1	3.41
$1.36 \cdot 2 \log_{10} N$	1	2.45
the pointwise minimum, each scaled	2	1.08
$8.61 \log_{10} E_{\max} + 0.90 \log_{10} N - 5.31$	3	0.19

858 **A purely linear statistic reproduces the whole dependence.** Across the $E_{\max}$ sweep PR_{delay}
859 rises over the same range as MG; across the record sweep it is flat; and a single logarithm in $E_{\max}$
860 fits it more tightly than any form fits anything else in this appendix. Most of the ceiling is therefore
861 the number of components the delay window resolves *linearly*.

862 **The unexplained tracking limit.** The ceiling lies near eleven components for no reason we can
863 give. It does not follow the measured effective rank, which equals r throughout, nor does any form
864 we fitted account for it.

865 The identifiability ratio of section 3.4 does register it. At $r = 20$ the ratio stays above the band at
866 every record length, so a longer record never makes an unidentifiable estimate identifiable; it falls
867 into the band as $E_{\max}$ grows, at about the rank where tracking stops.

868 M Two controls separating the estimator from roughness

869 *The number of components changes and the roughness does not (figure 3).* Both systems are a sum
870 of four sinusoids. In the first the four frequencies are rationally independent, so the path lies on
871 a four-dimensional torus and $d_{\text{act}} = 4$. In the second they are harmonics of one irrational base
872 frequency: a single phase fixes all four sinusoids at once, the path is a circle, and $d_{\text{act}} = 1$. The base
873 frequency of the second is set by bisection so that the two have the *same* roughness. The schedule
874 between them ramps the envelopes; every window a ramp reaches into is left unscored.

875 *The roughness changes and the number of components does not.* Five monotone observers $g(x) =$
876 $x + ax^p$ are applied to one four-phase system. Each is invertible, so it may bend the recorded set but
877 cannot add or remove a coordinate. Each also changes the local scale $g'(x)$ and with it the spacing of
878 consecutive samples.

Table 23: Both controls, median over eight seeds. Above the rule, two systems matched in roughness and differing in the number of independent phases. Below it, one four-phase system under five invertible maps. PR_{delay} is the linear participation ratio of the delay covariance, included because it follows the roughness rather than the active dimension in both halves.

	d_{act}	estimate	roughness	PR_{delay}
four independent phases	4	3.92	0.091	2.20
one phase, four harmonics	1	1.11	0.091	4.10
identity	4	3.92	0.091	2.20
$x + 0.25x^3$	4	3.96	0.098	2.32
$x + x^3$	4	3.96	0.110	2.58
$x + x^5$	4	4.01	0.146	3.61
$x + 0.1x^7$	4	4.02	0.166	4.28

879 The tolerance was fixed before the runs, at a tenth of a component about four. Table 23 shows it
880 holds.

881 Both controls run on a constructed system whose phases are known, so they say nothing about a
882 training log.

883 Two findings count against the estimator. The linear participation ratio is higher on the circle than on
884 the four-torus (table 23), so anyone treating it as a count of phases would order the two backwards.
885 On the circle the estimate is also not geometric: every neighbour surviving the exclusion there is a
886 sample from another pass around the circle. The local speed of the curve multiplies all of a point's
887 neighbour distances alike; speed cancels from the log-ratios of equation (3). The residue depends on
888 the arithmetic of the base frequency: two of the eight seeds drew a frequency whose multiples cluster
889 into near-duplicates; their estimate is reproducible from the fractional parts of those multiples alone,
890 with no signal and no neighbour search.

891 **N Compute cost and code availability**

892 **Compute cost.** Everything here runs in a few hours on one consumer GPU and one desktop CPU.
893 The largest GPU item is the edge-of-stability experiment of appendix K, which takes half an hour on
894 a single T4; the analysis is the more expensive half and runs entirely on CPU, a few core-hours per
895 experiment.

896 **Cost of the trajectory sketch.** At each logged step the sketch keeps 4096 floats against the full
897 parameter vectors of appendix J: a run whose raw trajectory would occupy 88 megabytes occupies
898 2.5. We train the same configuration with and without the sketch attached, alternating the order so
899 that warm-up cannot favour either. At a logging stride of ten steps the sketch adds about six per cent
900 to the training time. `sketch_cost.py` records the measurement.

901 **Code availability.** All code, configurations and result files behind every number and figure in this
902 paper will be released publicly on acceptance. The figures regenerate from the committed result files
903 with a single command.